





INVESTIGATION REPORT

MonWeb: a Real-Time Website Monitoring Alert System for Enhanced Website Security and Availability

BY

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ABSTRACT

Nowadays, most companies are using websites to promote their product and connect online with customers, so website security has become a more important and focused item for the technical team. There are a few challenges of the official website that the industry will need to face, such as the high-cost issue, limitations of error website detection, and being comfortable for non-technical users. The aim of this project is to develop a MonWeb system, a Real-Time Website Monitoring Alert System, to ensure the availability and security of the target websites. This project proposes a critical weakness by researching the existing systems, such as high price issues, less focus on non-technical users, and less detection of anomalies. The MonWeb will integrate artificial intelligence and natural language processing to increase the accuracy of website monitoring detection. It also provides user-friendly daily and weekly reports for non-technical users to gain an understanding of the detection reports. The system integrates Sustainable Development Goal, which is Industry, Innovation, and Infrastructure Sustainable Development Goal (SDG) 9, by focusing on building resilient infrastructure, fostering inclusivity, promoting sustainability, and encouraging innovation. This project supports SDG 9 because it will show continuous monitoring to build a structure for resilient digital infrastructure and increase inclusivity by creating a user-friendly interface and dashboard for non-technical users to understand the system progress and detect the results report.

Keywords: Artificial Intelligence, Website Monitoring, Natural Language Processing, Sustainable Development Goal 9， Website Detection

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CHAPTER 1: INTRODUCTION

1.1 Introduction

The digital age has made the internet a vital component of various industries, including businesses, media organizations, news organizations, and company homepages. Consequently, websites serve as a bridge between organizations and the users, the aim is to provide online services, e-commerce, news distribution, information sources, customer interaction, entertainment sharing, and product feedback. Therefore, websites have become one of the main ways for organizations to provide information and services, and the security and accessibility of websites have become increasingly important. However, if organizations fail to regularly monitor their target websites, many problems may arise, such as server failures, increased loading times, network outages, and increased risk of cyberattacks. Ultimately, transaction risks, customer losses, user trust, information loss, and service availability will all be affected by these problems.

Nowadays, in this technical environment, most companies will focus more on the availability and accessibility of the official website because the security of the website will impact the company's ability to connect with its customers. However, the traditional website monitoring technique can provide somewhat accurate results but requires high human resources and costs for frequent manual checking and testing. So, many of the website monitoring systems have been developed to meet the company's needs. The importance of a secure website is increasing, and the website knowledge and skills are also required by companies just to ensure that most of the technical team can understand how to use the website monitoring system, thereby making the non-technical team face big challenges in the detection results.

In order to guarantee that the project focuses on creating resilient infrastructure, promoting inclusion, advancing sustainability, and stimulating innovation, it will be in compliance with Sustainable Development Goal (SDG) 9: Industry, Innovation, and Infrastructure. The system developed by this project will provide a reliable website monitoring and alert system to ensure the reliability, security of the target website and organizational operational efficiency. In addition, this system is also able to support and benefit the cybersecurity team, IT

administrators, and organizations that are working on the company's official website, such as enhancing the users' awareness of potential threats.

The objective of this project is to maintain the monitoring websites and prevent the website's system performance by developing a real-time website monitoring and alert system, which is the MonWeb system. MonWeb system integrates HTTP(S), ICMP ping detection, and Artificial Intelligence (AI) techniques to implement the detection functions. For example, HTTP (S) and ICMP ping detection can help identify the target website status, identify errors responding, and network connection problems. Artificial Intelligence can provide an increased accuracy of detection result functions, to reduce the effect of wrong identification results. Not only that, but the MonWeb system can also provide generation detection result report functions to enable users to save the daily and weekly status reports of the target website. The daily and weekly website status report will integrate Natural Language Processing (NLP) to generate and ensure the report is simpler and easier to understand

1.2 Problem Background

Although immense progress has been achieved in website monitoring systems in the current era of real-time monitoring technology in comparison with the conventional method of monitoring websites, website security and accessibility are confronted with many challenges due to the increased dynamism and complexity in the applications being developed on the websites. This is closely linked with the Sustainable Development Goal SDG-9, which stresses the need for developing resilient infrastructure and promoting inclusive and sustainable industrialization along with innovation. The following points shall elaborate the challenges relating to website monitoring technology along with its association with SDG-9.

Problem 1: Absence of Real-Time Monitoring

Nonetheless, existing website monitoring systems are still based on traditional methods involving manual inspection, particularly checking the website manually. According to a research report by Koch & Superhero (2025), most companies are checking their websites manually at least 3-4 times in a day because they think this is enough to ensure normal operation of their websites. In fact, as revealed by Arshad (2025), in website monitoring systems, there is partiality in terms of monitoring. The thing is most website monitoring systems cannot monitor in real-time. For instance, most website monitoring systems cannot actually identify from the perspective of the user's actual experience, but instead monitor based on how websites respond normally on the internet from the perspective of the server. Therefore, this results in identification results that cannot show real accurate information from the perspective of the user's actual experience (Challenges in Website Monitoring | ManageEngine Applications Manager, 2025). What's more is that traditionally, as clarified by Schmitz (2020), the mentioned process cannot process massive amounts of data. It shows its inadequacy in providing proactive identification capabilities as well as identification solution schemes to its users. Therefore, this process gives birth to an extremely non-friendly interface because, as stated, this process cannot accurately identify identification results. Many alerts are generated so they need to spend much time as well as manpower in troubleshooting as well as in pinpointing this problem until this may stop this team in pointing out major concerns. It is rather non-compliant with Sustainable Development Goal 9.

Problem 2: Lack of Real-Time Alerts and User Knowledge

Most website monitoring systems also have several detection functions and tools but cannot offer the capability of sending alerts in real time. In fact, the knowledge of users regarding alerts is also limited, making it impossible to create an alert setting for real-time detection. For instance, website monitoring systems are supposed to identify website anomalies with regard to response and runtime, after which the decision to send alerts to the team to act accordingly would be made. The delay in sending alerts also impacts the team, which would not be able to act immediately due to the potential danger, subsequently hampering the sustainable development of the required SDG 9 to enable the user to get the alerts in real time.

Problem 3: Inability to Effectively Handle Complex Websites Indeed

In the current technologically advanced environment, websites are filled with third-party functionalities, APIs, and dynamic content. This is considered challenging for website monitoring systems, which become inefficient due to these factors. For instance, online shopping, social networks, and websites with interactivity turn static websites into complex platforms with diverse dynamic content. This is considered one of the reasons that the potential threats of the websites are hidden. The fixed pages of several website-monitoring programs, particularly the homepage of prominent websites, are monitored. This is considered misleading information regarding detection. However, the homepage is an entrance point that is not the entire experience provided by the users. For instance, checkout and user profiles are the pages that are considerably the most deserving of being monitored. This constrains the sustainability objectives of Sustainable Development Goal 9 due to the lack of continuous improvement of digital infrastructure.

Problem 4: Overemphasis on High Quality and Complexity of Technology Ignores Assistance to Non-Technical Users

The current website monitoring tools place too much emphasis on the high quality of the results and completeness, utilizing different professional functions and complex interfaces. Consequently, the results of the analysis are presented to users in raw codes, data, and findings, thereby posing difficulties to non-technical personnel to comprehend the results. As stated by [reference needed], a high-quality user interface is enough to cater to the needs of different

personnel in an organization; that is, analysts require processed results and personnel in charge of running the organization require correct alerts and corresponding problem-creating sources. Next, when the technical team is making a decision to solve the issues, the technical team are making different decisions for a deeper understanding. It will cause the technical team to misfocus on the key problem. A high-quality technique system will become a challenge for implementing the inclusion and innovation agenda of Sustainable Development Goal 9.

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1.3 Project Aim

To develop a real-time website monitoring and alert system using Artificial Intelligence and Natural Language Processing to ensure users can continuously track target websites and obtain information on their availability and status, thereby enhancing user awareness and website security.

1.4 Objectives

Objective:

1. To design and develop a real-time website monitoring alert system that uses artificial intelligence to improve detection accuracy and ensure website availability.

2. To integrate natural language processing (NLP) into the system to generate user-friendly daily and weekly reports, enabling non-technical users to understand website monitoring results

3. To implement an automated alert mechanism to notify users when abnormal activity or security issues are detected.

4. To design a user-friendly visualization feature that allows users to understand the real-time status of target websites. To provide customized website management features that allow users to manually add and remove target websites.

1.5 Scope

The objective of the project will be to design an internet monitoring system to supervise and report any accessibility, slow accessibility, or down time experienced by a target site as may be inputted by the user. This will be achieved through the use of Artificial Intelligence and Natural Language Processing algorithms to offer effective suggestions and reporting when accessibility or down time is detected.

Tasks to be Executed:

- Design and implement a user-friendly interface that allows users to intuitively understand the website status, such as response time graphs, the number of alert notifications, and the number of abnormal notifications.
- By using natural language processing (NLP) technology, develop user-friendly daily and weekly target website status detection results.
- Develop an AI-driven backend system focused on improving the accuracy of detection results and the effectiveness of processing target website data.
- Implement an automatic alert mechanism to send alert notifications, such as pop-ups, when anomalies, abnormal responses, or abnormal statuses occur on target websites.

Constraints:

- Due to resource constraints, the system's AI and NLP capabilities will rely on pre-trained models and publicly available datasets.
- Besides monitoring target websites and providing alert mechanisms, the system does not include advanced anti-tracking or automatic remediation features.
- The system will be presented to users on a web platform and can be accessed via a browser.
- Due to the limitation time, the system will prioritize detection accuracy and functional completeness, rather than relying on specialized enterprise-grade infrastructure.

Inclusions:

- The system will incorporate artificial intelligence algorithms to analyse monitoring results, improving detection accuracy, and ensuring accurate alerts.
- The system will use Natural Language Processing (NLP) to generate user-friendly detection results, target website status, and response times.
- A concise user interface includes intuitive detection results, response time graphs, and historical data for easy interpretation of the data.

Exclusions:

- This project will not involve developing mobile applications or using enterprise-grade infrastructure for target website monitoring.
- This project will use publicly available data for development and testing and will not include real company data.

1.6 Potential Benefit

Tangible Benefits:

- Saves Manpower and Costs: The system employs automated detection and alerting mechanisms to reduce manual labour and frequent manual checks.
- Enhances Website Availability: The system monitors website status in real time, reducing website anomalies and downtime.
- Saves Time: The system uses a user-friendly interface and generates user-centric detection reports, ensuring users can quickly resolve website status issues.

Intangible Benefits:

- Enhances User Cybersecurity Awareness: The system can potentially cultivate users' awareness of website security and proactively prevent potential cyber threats.
- Improves User Experience: Artificial intelligence and Natural Language Processing (NLP) enhance the user experience, ensuring a smooth interaction with the system.
- Enhances Organizational Reputation: An organization's emphasis on website security can increase customer trust in the organization's website.
- Improves Team Communication: The user-friendly interface allows non-technical personnel to easily interpret detection results, reducing communication barriers and promoting teamwork.

Targeted Users:

The system is designed for cybersecurity teams, IT administrators, and organizations that manage commercial websites. It will be deployed as a web-based application to allow the target audience to easily add and monitor websites, and to help the target audience quickly identify website response and potential network threats, thereby increasing the availability and security of the website.

1.7 Overview of the IR

The purpose of this report is to present the basic operation and planning of the MonWeb system with alignment to the Sustainable Development Goals (SDG 12) because the MonWeb system helps to remind users of the monitoring of the website in a real-time manner, which aims to ensure the security of the website itself. The research reported here will be conducted from the theoretical and technical perspectives.

Chapter 1: Introduction mainly highlights the background and results of the project, the goals, strategies, advantages, and project planning. Chapter 1 explains how the project is linked to the achievement of the target of the SDG 12, specifically focusing on the issue of digital infrastructure security.

In chapter 2, the topic of focus is the review of literature, mainly dealing with the study of existing technologies that monitor websites, artificial intelligence, natural language processing, and others. The review of literature studies the existing technology and theoretical studies that help improve the MonWeb system, dealing with its shortcomings.

Chapter 3 mentions the technical methods adopted in the design and implementation phase of the project. This encompasses methods for data collection and design, plans for system development, data analysis methods, and user feedback collection by means of surveys. Acquiring focused knowledge about the process of system development helps us address user needs and expectations better.

This last chapter summarizes the findings of the research contained in the report and the shortcomings that exist in the research field. This chapter also evaluates the sufficiency of the conducted survey that should be adequate enough to fulfill the accomplishment of the project goals and objectives. This report is detailed enough to provide clarity at the level of strategic planning.

1.8 Project Plan

Investigation Report				
Task	Duration	Start	End	Status
Project Proposal Form	6 Days	21/09/2025	26/09/2025	Completed
Chapter 1: Introduction				
Introduction	1 Days	28/09/2025	28/09/2025	Completed
Problem Background	2 Days	29/09/2025	30/09/2025	Completed
Project Aim	1 Days	1/10/2025	1/10/2025	Completed
Objectives	2 Days	2/10/2025	3/10/2025	Completed
Scope	2 Days	4/10/2025	5/10/2025	Completed
Potential Benefit	2 Days	6/10/2025	7/10/2025	Completed
Overview of IR	3 Days	8/10/2025	10/10/2025	Completed
Project Plan	3 Days	11/10/2025	13/10/2025	Completed
Chapter 2: Literature Review				
Domain Research	18 Days	14/10/2025	31/10/2025	Completed
Similar Systems	9 Days	1/11/2025	9/11/2025	Completed
Technical Research	14 Days	10/11/2025	23/11/2025	Completed
Chapter 3: Technical Research				
System Development Methodology	8 Days	24/11/2025	1/12/2025	Completed
Data Gathering Design	6 Days	2/12/2025	7/12/2025	Completed
Analysis	5 Days	7/12/2025	11/12/2025	Completed
Chapter 4: Conclusion				
Conclusion	2 Days	12/12/2025	13/12/2025	Completed
References& Appendices	2 Days	13/12/2025	14/12/2025	Completed
Final Checking	3 Days	15/12/2025	17/12/2025	Completed

CHAPTER 2: LITERATURE REVIEW

2.1 Domain Research

2.1.1 Artificial Intelligence (AI)

Artificial intelligence (AI) refers to the use of programmed computer controlled and digital computer-controlled robotic systems to perform tasks resembling those of human professionals, such as chatbots, recognition engines, medical advice, computer search engines, and speech recognition (Muhammad Tuhin, 2025). Modern AI has seen tremendous advancements in computer processing speed and memory capacity, possessing powerful algorithms capable of performing tasks related to intelligence, including creating original images, videos, text, driving systems, backend data analysis, and frontend chatbots and recognition robots (Craig et al., 2024). AI encompasses various methods of simulating human learning and algorithms for autonomous problem-solving, and includes various subfields such as deep learning, natural language processing, and machine learning.

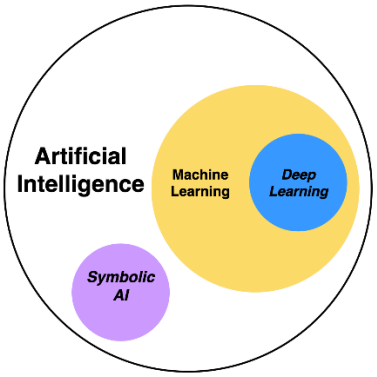


Figure 1: Artificial Intelligence (AI)(builder.aws.com)

Within the technological and scientific fields of AI, machine learning is one of the most important branches because its algorithms and techniques can learn and analyse automatically without explicit programming, thus improving upon the results (Sha, 2025). Machine learning techniques and algorithms are applicable to various types of challenges and data, such as logistic regression, decision trees, random forests, support vector machines (SVM), k-nearest neighbours (KNN), clustering, and one of the most popular algorithms today: neural networks (Virellan, 2025). Neural network algorithms mimic the structure and concepts of the human

brain to achieve interconnected components the neurons in the human brain. The benefits of this approach are tangible, as complex data analysis and processing can be accomplished through node layers (Rastogi, 2025). Machine learning can be broadly categorized into three types: supervised learning, unsupervised learning, and reinforcement learning.

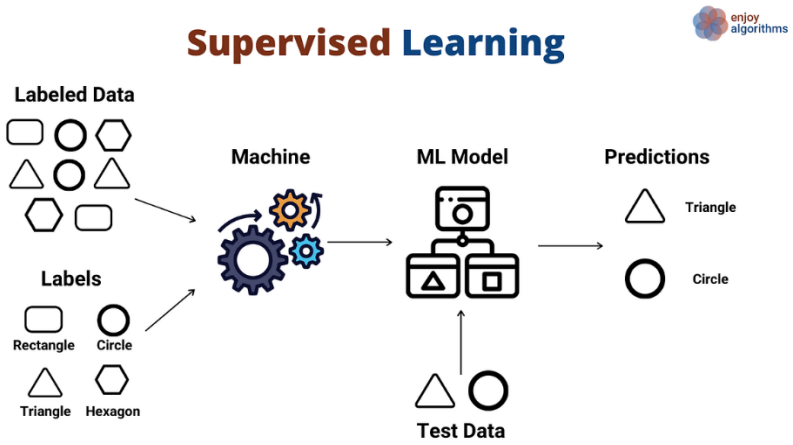


Figure 2: Supervised Learning(medium.com)

Supervised learning involves training a model on a labelled dataset. This means the labels used are unrestricted values, ranging from class labels to real numbers (Liu & Wu, 2012). Therefore, during training, the model learns the correlation between inputs and outputs, enabling it to predict new or previously unseen outputs based on new inputs. For example, supervised learning algorithms are commonly used in classification problems such as logistic regression, decision trees, and support vector machines; image recognition is one example (Tiwari, 2022). When using supervised learning to recognize images, each image in the dataset is labelled, for example, as a cat or a dog. Supervised learning algorithms can accurately identify and classify each image (Revanth, 2025).

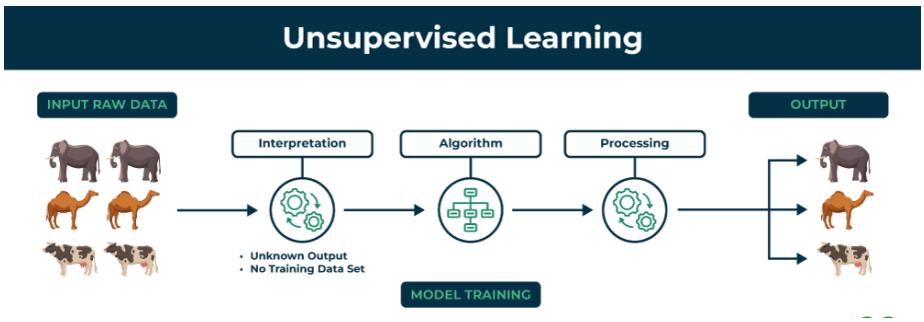


Figure 3: Unsupervised Learning (mindlabinc.ca)

Unsupervised learning, on the other hand, refers to training a model on an unlabelled dataset. This involves analysing unlabelled data to discover hidden patterns or groupings (J E T Akinsola, 2017). Therefore, this learning method is mainly used for clustering or dimensionality reduction. For example, clustering groups similar data points together, while dimensionality reduction obtains the main components of the groups to reduce the number of random variables to consider (Saiganesh Ramesetti, 2020). A common use of unsupervised learning algorithms is to segment business customer groups specifically by analysing purchasing patterns and demographic data. Thus, supervised learning develops predictive models based on input and output data, while unsupervised learning groups and interprets data based on input data (Meena & Jain, 2019).

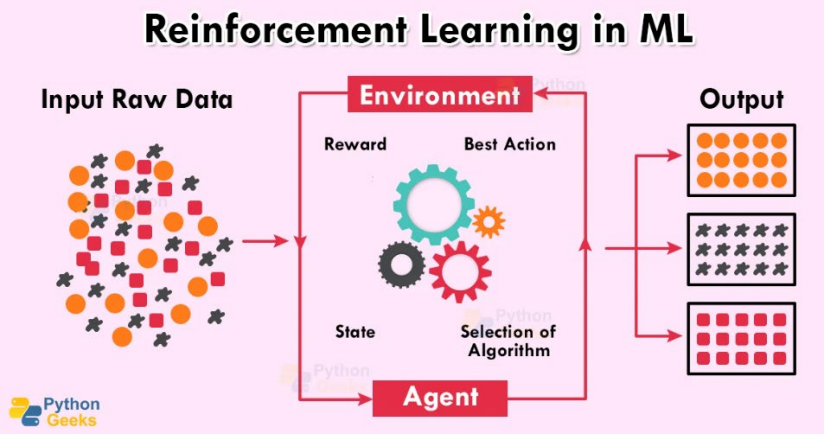


Figure 4: Reinforcement Learning (analyticsvidhya.com)

The last core type of machine learning is reinforcement learning. This type is mainly reflected in decision-making tasks because reinforcement learning involves continuous trial and error and receiving feedback to obtain the best reward (Murphy, 2024). Therefore, each decision made in reinforcement learning is constantly changing due to rewards and penalties, thus affecting the reward outcome (Kommey et al., 2024). Common examples of reinforcement learning are games, robots, and resources (Memarian & Doleck, 2024) (Riedmann et al., 2025).

With current technologies and algorithms, machine learning has become essential for enhancing cybersecurity. For example, machine learning is a major source of technology for enhancing the intelligence and automation of website monitoring systems. Supervised learning can use daily website detection results and previous detection results for labelling and analysis to predict the likelihood of future website anomalies. Meanwhile, machine learning offers the

advantage of continuous improvement, allowing it to constantly adapt to the cybersecurity environment and adjust accordingly, thus enabling it to make highly accurate decisions. Intelligent systems enhance the detection accuracy of website monitoring systems, thereby increasing user trust, and providing better technical support.

2.1.2 Natural Language Processing (NLP)

Natural Language Processing (NLP) is a branch of artificial intelligence, computational etymology, and human linguistics. NLP exists to enable computers to understand human language expressions, such as those expressed by computers (Rayhan et al., 2023). Alternatively, NLP is an algorithm that uses language processing rules and language models to process human language into numbers and codes that computers can understand (Supriyono et al., 2024).

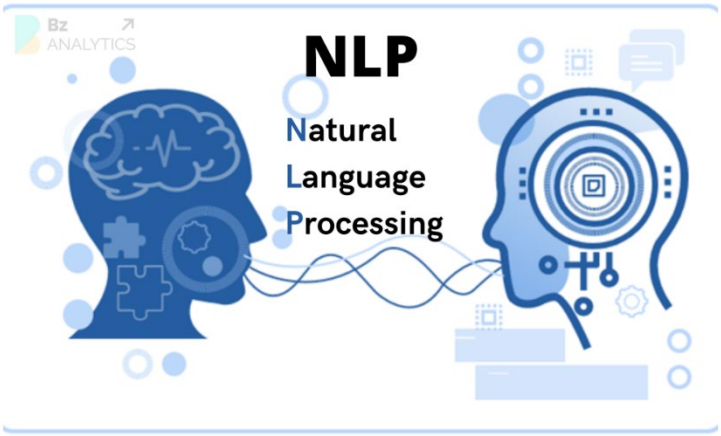


Figure 5: Natural Language Processing (NLP) (parascript.com)

Natural Language Processing (NLP) originated from Alan Turing's paper on computers and intelligence, also known as the Turing Test. At that time, in 1950, although there were already articles on NLP, it was still an undeveloped technology, and very few people understood its existence (Das & Das, 2024). From 1950 to 1990, a series of specifically designed rules for NLP began to emerge, such as language rules, information rules, and grammar rules. Subsequently, researchers shifted from manually creating rules for NLP to allowing it to learn from real-world data, thereby improving automated learning techniques (Simplilearn, 2024). Therefore, the rapid rise of the internet and online data in 2000 led to

numerous research articles on Natural Language Processing (NLP), including NLP algorithms, unsupervised learning algorithms for artificial intelligence, and semi-supervised learning algorithms (Supriyono et al., 2024). This learning algorithm technology allows NLP to extract language rules from massive amounts of internet data and identify patterns, continuously addressing algorithmic shortcomings. Today, NLP has become an advanced technological achievement, enabling computers to express language in a natural way to users. NLP has been used to develop many everyday products, including Google Translate, Google Assistant, online chat bots, Bing Search, Chatgpt, Gemini, and the YouTube platform.

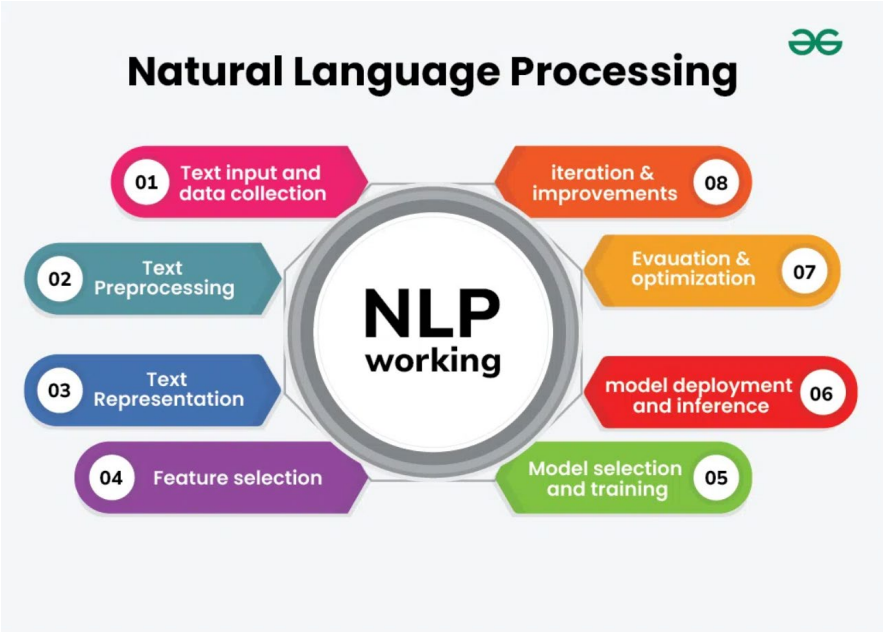


Figure 6: Workflow of Natural Language Processing (NLP)(geeksforgeeks.org)

Natural Language Processing (NLP) possesses a complete algorithmic approach to analysing and processing human language and literature, breaking it down into machine-readable data. Therefore, NLP can be broadly categorized into two types: Natural Language Understanding (NLU) and Natural Language Generation (NLG) (Hegde, 2025). NLU's role is to extract information from human text or speech and analyse it to derive machine-understandable meaning, regardless of grammatical or pronunciation errors. Thus, NLU is crucial for human-machine communication; it serves as a bridge for effective dialogue. Natural Language Generation (NLG), on the other hand, uses the data analysed by NLU to create natural language that humans and robots can mutually understand (Rayhan et al., 2023).

Consequently, many organizations in the current market leverage NLG's capabilities for organizational benefits, such as human resources and information technology departments.

The figure shows the workflow of Natural Language Processing, which includes text input and data collection, text pre-processing, text representation, feature selection, model selection and training, model deployment and inference, evaluation and optimization and iteration and improvements (Rayhan et al., 2023). At the first step, which is text input and data collection, it is used to prepare all the data that the machine needs. So that the machine can collect all the language data prepared for step two and use the language data to train language models (Zhang & Zhang, 2025). Next, step two is to process the language data because unprocessed language data is chaotic; this step requires breaking down and cleaning all the data into regular language. For example, on this step, it needs to remove all the characters and irregular symbols, which are ",", ":", "?", "-", "!", "@", "#" and "*".

After that, step three is text representation, which means converting the human language to machine language. Transformer embedding model is one of the functions NLP will use to process the text into corresponding numerical. The transformer embedding model can also process audio, images, and graph data. This step is the most important because it is the key bridge to connect the human language to machine algorithms (Luke et al., 2023). After all the text conversion into a number variable, feature Selection will be processed to get the needed number variable from the number data because not all numerical variables are needed by users, and some can even mislead NLP processing. Therefore, the goal of this step is to allow the model to concentrate on the necessary data and reduce resource consumption, such as the fewer the number of variable features, the more efficient the efficiency process algorithm (Liu et al., 2025).

Afterwards, the step of model selection and training is for NLP to absorb and improve the algorithm model. From this step, NLP can select a suitable model and algorithm for processing the language data, such as Naive Bayes, Support Vector Machine, Long Short-Term Memory Network, and Recurrent Neural Networks (Rayhan et al., 2023). model has been trained, it will be deployed to the platform to provide services to the users, which is the step of model deployment and inference. In this step, the model needs to be deployed to perform the task provided by the users (Khan et al., 2024).

Not only that, to verify the effectiveness of NLP results provided by users, it is necessary to continuously monitor NLP performance, such as accuracy, precision, and F1 score, which is a step of evaluation and optimization (Antonin Bergeaud et al., 2025). The focus of this step is to check whether NLP is giving users incorrect results, analyse the source of the errors and their impact, and use the analysis results to help the model develop a strategy for continuous improvement to obtain a high-quality model. This step is used to keep collecting the data from the user because NLP can use the data to retrain the model and the process method. It can help to update and upgrade the model by getting the data from users (None Supriyono et al., 2024).

2.1.3 Website Monitoring

Website monitoring refers to the use of advanced technologies to effectively detect the security, availability, and suitability of a target website (Akamai, 2024). Uptime, response time, performance, and possible vulnerabilities are primary data used in monitoring the security and availability of a target website. Moreover, the most crucial function of website monitoring is to guarantee that the target website is online in real-time. For instance, when users access a company's official website, they will get the information they need accurately and safely. On the contrary, if users use the target website through some activities like payment pages, loading pages, and dashboards, and receive malfunctions or suspected potential threats, website monitoring should send error codes and information in time to corresponding technicians. Therefore, a website is an essential channel for communication between business and users (Achie Barret, 2023). The website monitor tools are used and traced by an organization's internal technical department, including website administrators, IT professionals, network technicians, and front-end and back-end developers. Since they know the basic knowledge of networks and some technical skills, by sending alerts to them when website monitoring detects faults and letting them perform suitable strategies as soon as possible will hugely reduce resource consumption and enhance user trust on the websites (Hariyanto et al., 2025).

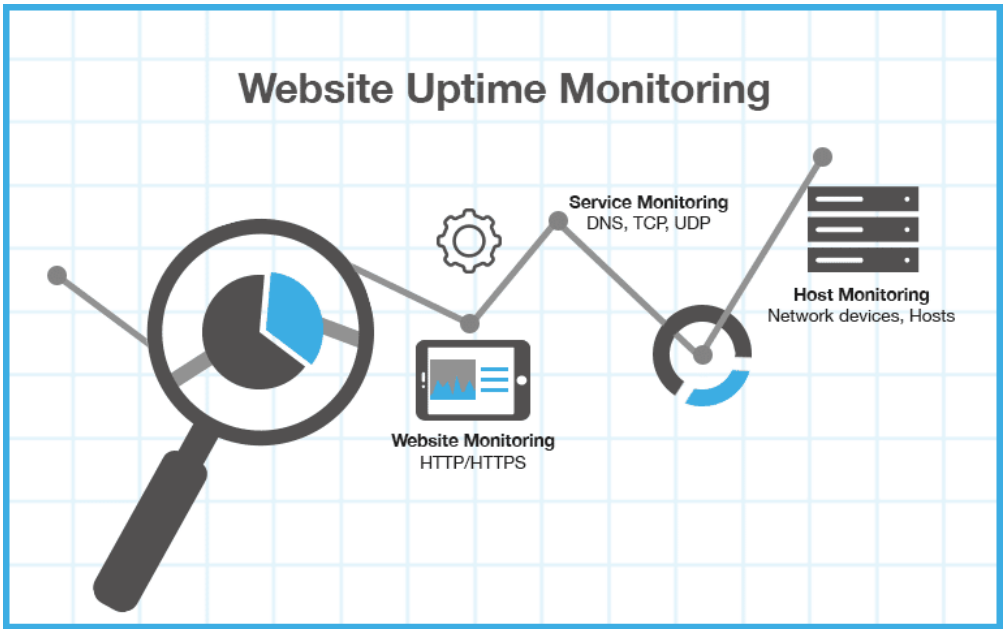


Figure 7: Website Uptime Monitoring (100kresults.com)

With rapid development and wide usage of the internet, especially when disseminating news (O'Brien, 2024). Newspapers are no longer the only source of information and market analysis. Instead, people depend on different social media platforms and news provided online. Some of the representative online news include The Star, Sin Chew Daily Malaysia News, Oriental Daily, China Press, Google News, and The Malaysian Mail. Corporate websites have, therefore, become a necessary tool for every business, an expensive affair in terms of resources to ensure the website is accessible (Srushti Vachhrajani, 2025). Website monitoring allows a better user experience, is more accessible, improves cybersecurity, and ensures continuity of the business.

It improves the user experience in website monitoring due to the fact that website performance provides information about the performance of a website; an organization is able to know the capabilities of their website. The performance of the website determines the loading of the page's speed and user navigation (Admin, 2024). Website loading times or other performance issues can be found that cause pages to fail by tracking website performance. This significance transforms into continuous satisfaction by users and, finally, retention of customers. Moreover, website monitoring can find website uptime in real time. This ensures that when a website fails to load or users cannot access important product-related websites, this creates massive economic losses regarding customers for the organization (Kannan et al.,

2024). Therefore, having an on standby professional website technical team helps in handling alerts related to website server problems together with abnormal website uptime, thereby reducing time wasted on website loading failure.

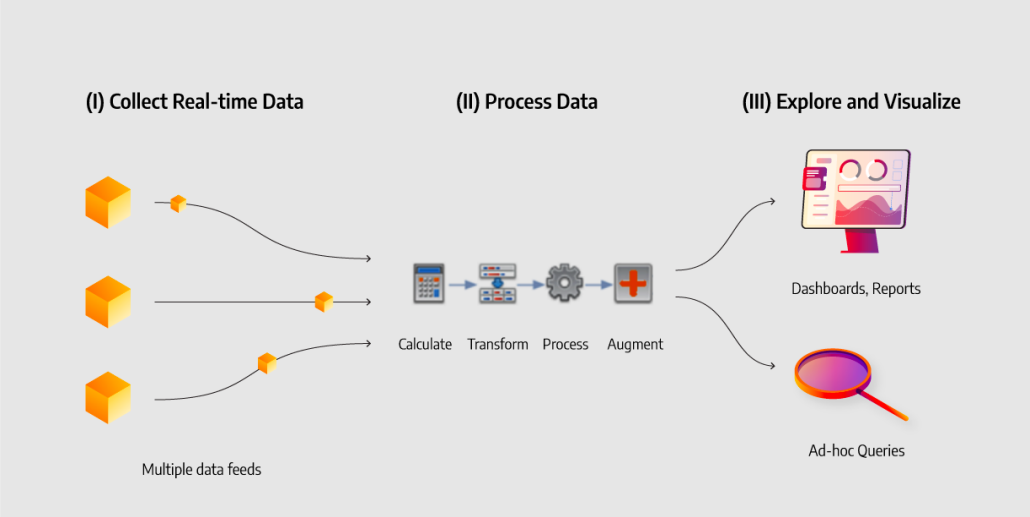


Figure 8: Analysis Real-time Data (amazon.com)

Real-time website monitoring involves comprehensive and detailed website security and operational information. The security and performance information provided by website monitoring enables organizations to produce appropriate responses to counter cyberattacks (García et al., 2023). Moreover, sophisticated cybercriminals today use data breaches and malicious tools as common tactics that expose websites continuously to the possibility of losses from cyberattacks (Charanjeet Dadiyala et al., 2024). For instance, outdated software, invalid security certificates, or suspicious activity are some of the aspects that lead to cyber threats greatly impacting website security and increasing risks. A secure website ensures that business continuity can be guaranteed to users, whether for important pages such as payment pages, company financial pages, or user interaction pages (Sarwar et al., 2022). Many financial users use trading websites to conduct business and therefore need sufficient protection so that customer transaction disruptions can be minimized when using these trading websites.



Figure 9: Active and Passive Monitoring (extnoc.com)

There are two types of website monitoring: passive and active monitoring. Passive and active monitoring are popular enterprise methods, including Infrastructure and Application Performance Monitoring (APM) (Andreozzi et al., 2025). Passive monitoring collects real-time website access data, making it impossible to obtain data and detect potential problems when there are no users accessing the website. An example of passive monitoring is the website monitoring system silently recording and observing all website activities and dynamics, such as daily website operation time and various data (Matthews et al., 2001). Active monitoring, on the other hand, simulates user website activity without requiring users to actively access the website (Andreozzi et al., 2007). This method allows for proactive detection of problems before users discover them. An example of active monitoring is that the website monitoring system acts as a virtual scout, constantly testing and observing the website's actions, and taking proactive actions such as interacting with the website and simulating normal user behaviour (Popp et al., 2024). Below are the differences of passive monitoring and active monitoring:

Table 1: Comparison of Active Monitoring and Passive Monitoring

Type	Active Monitoring	Passive Monitoring
Property of Data	Synthetic Data	Real Data
Quantity of Data	Small	Large
Requirements of Storage	Small Storage Requirements	Large Storage and Scalable Devices
Requirements of Calculate	Low Data Volume and Complexity	Large Data Volume and Complexity
Privacy Data Problem	Low Privacy Data Issue	High Privacy Data Issue

The property of active monitoring is Synthetic Data because all the data is simulated from the same type, such as a browser. It can ensure the data is regular. The property of passive monitoring is Real Data because all the data is collected from real users and real devices. It will make the data more complex and uncertain. Next, the quantity of data active monitoring is small because the monitoring frequency is controllable and limited. The quantity of data from passive monitoring is large because the monitoring frequency is controlled by real users and traffic; therefore, it is uncontrollable. Not only that, since the quantity of data from active monitoring is small, the storage requirements are also small. Passive monitoring needs a large storage requirement and scalable storage devices to store data; otherwise, the search speed will be slow. The calculation requirements of active monitoring are low because it needs a simple data structure and volume for completed analysis data. Passive monitoring needs a large calculation requirement since it has highly complex data that needs to be cleaned, filtered, and analyzed to get valuable information. Lastly, active monitoring has a low data privacy issue because the data are synthetic data and pose less privacy risk. Since passive monitoring uses real data from real users, it will include a high privacy risk.

2.2 Similar System/Works

2.2.1 Pingdom

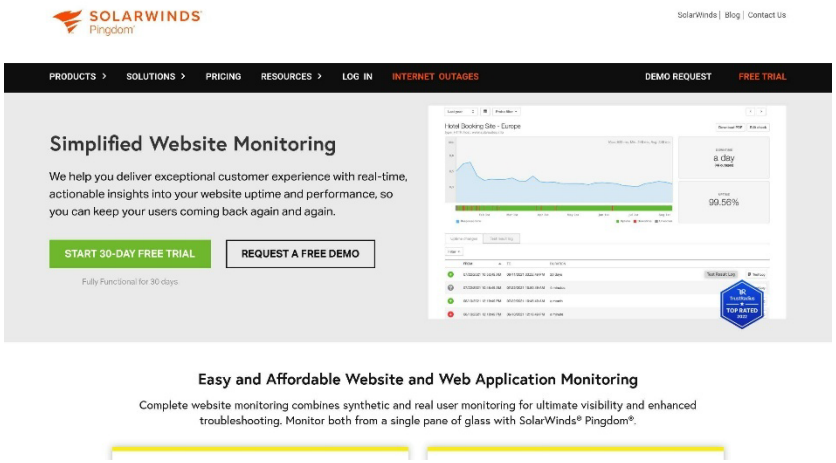


Figure 10: Pingdom Official Website (pingdom.com)

Pingdom is a website monitoring and performance optimization tool service company, founded in Stockholm and acquired by SolarWinds, based in Austin, Texas. In 2007, Pingdom was established and became one of the website monitoring solutions options for users and businesses on the internet, which include Spotify, Pinterest, Facebook, Twitter, Zendesk and MailChimp (Scott Charles, 2025). The target services of Pingdom are more than 100 locations in Europe, the Americas, and the Asia Pacific region, as the aim of this is to expand Pingdom's detection to global and become a full comprehensive website performance platform (Chowdhary, 2025).

The approach of SolarWinds, which is Pingdom's company, solves complex issues in a simple way. This approach helps Pingdom get trust from a lot of users and businesses of every size in every industry across the globe. It can be more productive for helping people and businesses. The purpose of Pingdom is to enrich the lives of the people, which include employees, customers, shareholders, partners, and communities (Venelin K, 2025). The mission of Pingdom is to provide simple, powerful, helpful, and secure solutions for customers.

2.2.1.1 Features

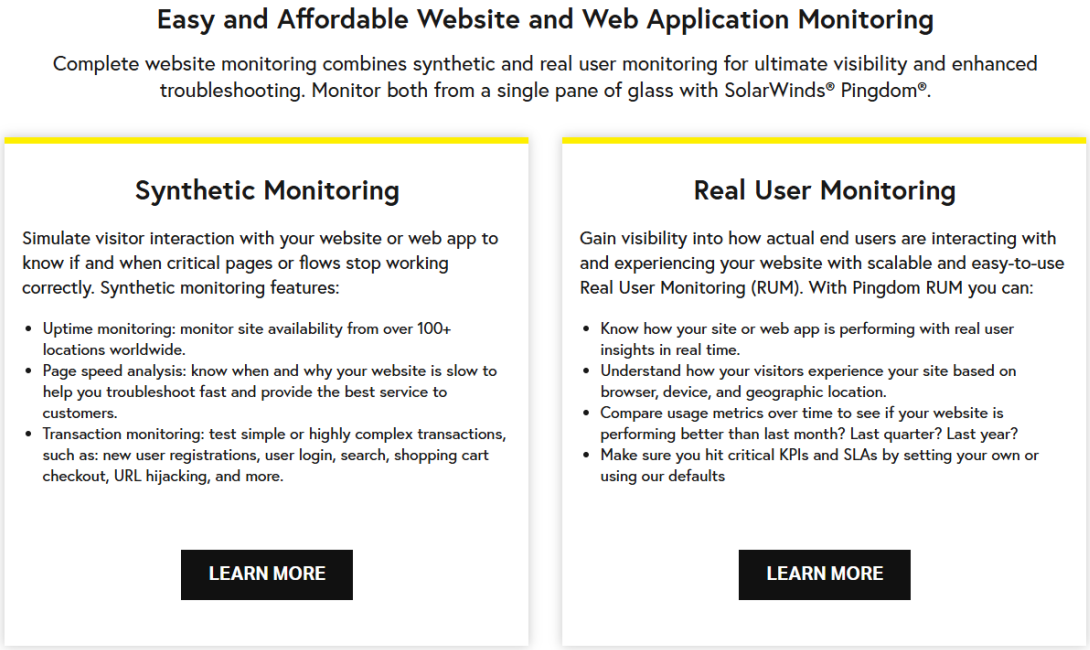


Figure 11: Types of Pingdom's Features (pingdom.com)

By achieving visibility and robust troubleshooting capabilities, Pingdom has combined two techniques, which are synthetic monitoring and real user monitoring. These technologies can provide detection and a solution which is suitable for users and businesses to ensure website accessibility and security even in complex network environments. The primary purpose of synthetic monitoring is to obtain website performance and information by simulating user behavior tests, while real user monitoring is through a tracking code to collect actual website data and behavioral information from real users accessing the website.

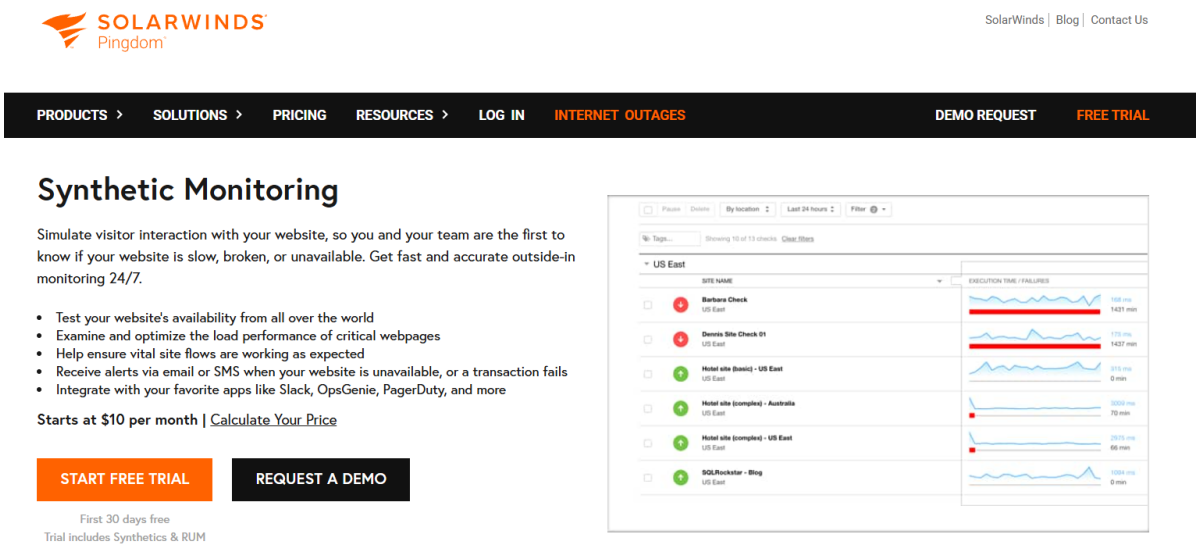


Figure 12: Synthetic Monitoring Features of Pingdom (pingdom.com)

Synthetic Monitoring

The synthetic monitoring can provide functions and results to help users and businesses, such as testing the target website's availability, detecting the load performance of critical webpages, ensuring the vital site flows are expected to work, and users can receive an alert message when the target website is unavailable.

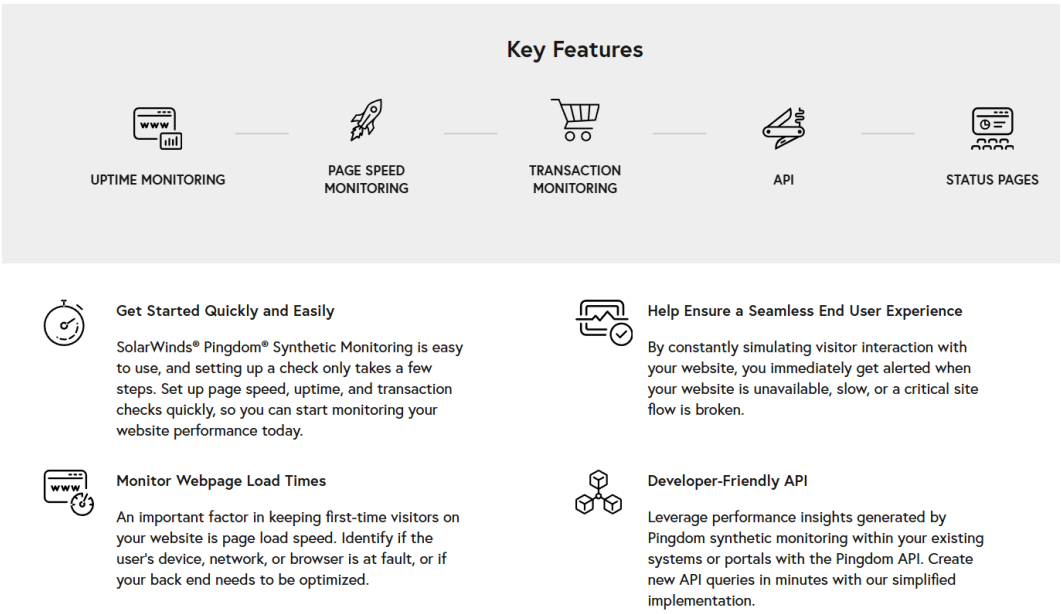


Figure 13: Key Synthetic Monitoring Features of Pingdom (pingdom.com)

The key features of synthetic monitoring in Pingdom are uptime monitoring, page speed monitoring, transaction monitoring, API, and status pages. When users face challenges with the target website, the solution of simulating user interaction with the website can alert the user, thereby improving the user experience when using Pingdom.

By using the synthetic monitoring, it can provide some benefits for users, such as getting the status of the website when it becomes unavailable, optimizing the user experience by increasing the load times, fixing the critical site flow problem before affecting business, using the Pingdom API to improve the website's performance and being able to share the performance against service level objectives. A high-quality alert system can help customers reduce the effect of the website going down. It can notify customers by email or SMS message, but the maximum alert messages are up to four times. Not only that, but granular webpage load test tools can also enable users to monitor the performance of every page element, which can identify the reason for hurting the web performance and track page speed history.

Next, some important websites should use synthetic monitoring to ensure they do not affect the business before an error appears or goes down. The example of important websites includes the login page, search page, filter page, checkout page and top-up page. A good detection period can alert the users when critical site flows stop working. In addition, Pingdom provides an API interaction to increase the website's performance. Pingdom's API is based on RESTful and HTTP, thereby users can create scripts or applications with similar functions to the Pingdom control panel. Users can also provide third-party proof to their customers by using the Share Performance Against Service Level Objectives function. These functions can help users to have a simple way to show and configure the pages to suit the brands.

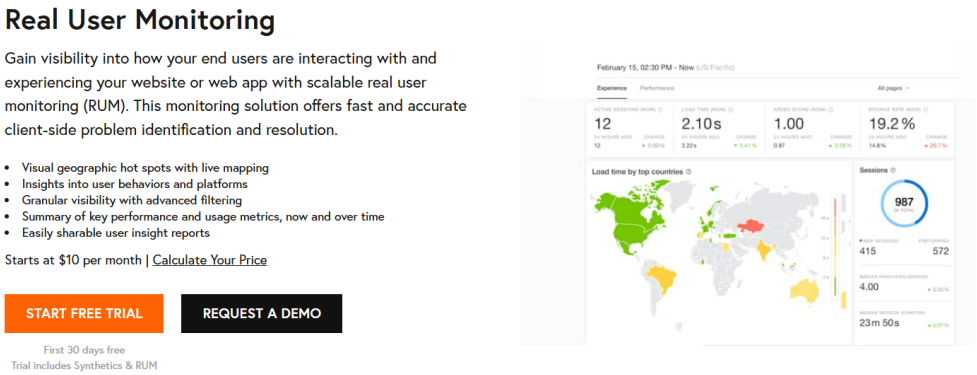


Figure 14:Real User Monitoring Features of Pingdom (pingdom.com)

Real User Monitoring

Real user monitoring can provide functions and results to help users and businesses, such as visual geographic hot spots with live mapping, identify insights into user behaviors and platforms, granular visibility with advanced filtering, a summary of key performance and usage metrics and easily share the detection reports.

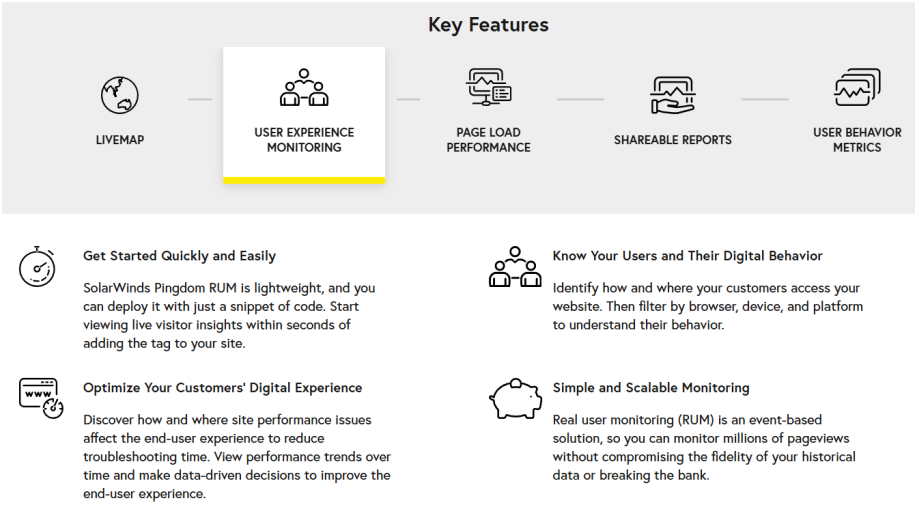


Figure 15: Key Real User Monitoring Features of Pingdom (pingdom.com)

The key features of real user monitoring are live map, user experience monitoring, page load performance, shareable reports, and user behavior metrics. Real user monitoring can provide and ensure the detection data are real, thereby Pingdom can based on the real data to understand the behavior. It can also continuously show the website's performance, which will help technical teams to shorten the troubleshooting time.

By using real user monitoring, it can provide some benefits for users, such as knowing users and digital behavior, identifying bottlenecks with advanced filtering, troubleshooting performance issues from the end user perspective, sharing and customizing reports on website performance and discovering and comparing usage metrics over time. According to the real user data, Pingdom can get the results about the reasons that affect the user experiences and adjust the website configuration accordingly. Next, troubleshoot the website's performance problem by monitoring the health and specific information of the website from the end user's perspective.

2.2.1.2 Strength and Weakness

Strength

The strength of Pingdom is trusted website monitoring results, full information and performance of the detection report, diversity of global monitoring sites, and diversity of alert systems. Pingdom can detect the website's status and based on the level of detection, notify the user using email or SMS messages in detection tests and user feedback (Singh, 2024). The accuracy of detection is important because a false alarm will provide a wrong detection message to the technical team. It will affect the loss of time for the technical team. Furthermore, a specific detection report can help the technical team save time on the filter process to get a high-quality experience (Scott Charles, 2025). The detection report will include macro-level page load times and performance levels, such as the micro-level, visually showing the loading order and time of each page's resources.

Weakness

The weakness of Pingdom is the high cost of using the Pingdom system, a lack of intelligent analysis, and high technical knowledge and skills requirements, which are not user-friendly for non-technical users. The best website monitoring tools for a company are effective and suitable cost (Singh, 2024). While Pingdom provides a free plan, it also has limited functions, such as monitoring one site, a limited number of target websites, and a limited number of notifications. In addition, user communication is one of the weaknesses when users face a challenge using Pingdom (Scott Charles, 2025). This is because Pingdom is a traditional website monitoring tool; it lacks an artificial intelligence function for analysis and provides suggestions to users. It will lead to a situation where users receive a notification alarm from Pingdom, users need to have enough knowledge and skill to understand the detection report and solve the website errors.

2.2.2 Datadog

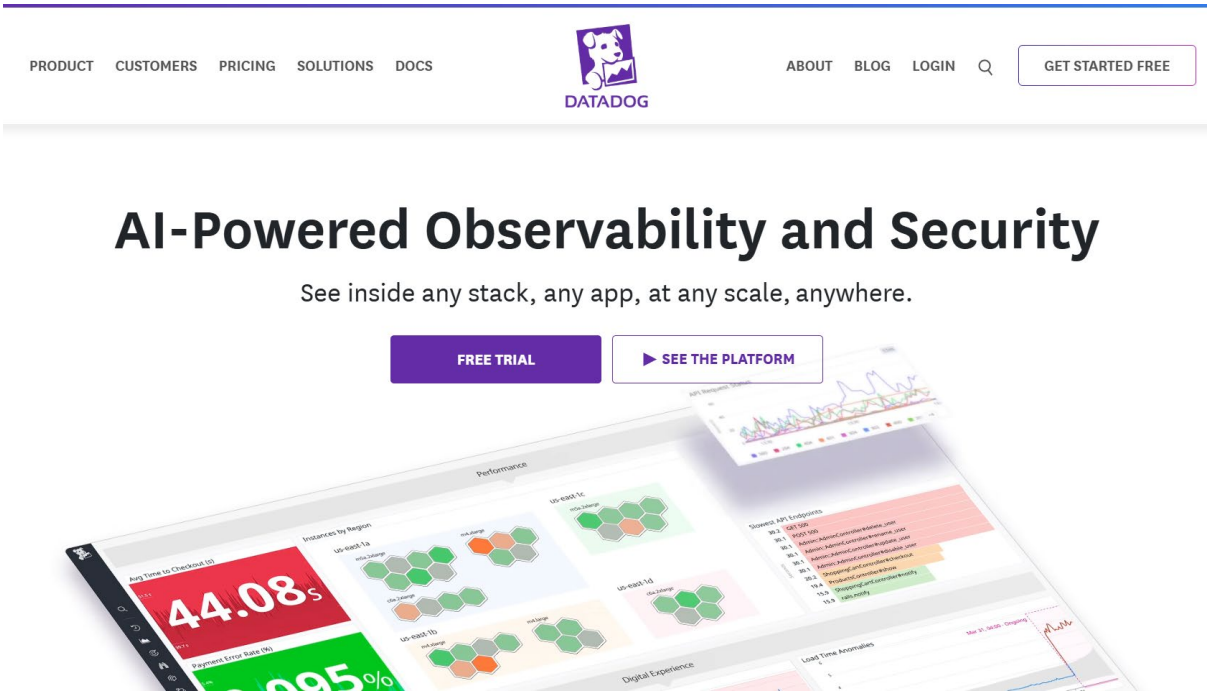


Figure 16: Datadog Official Website (datadog.com)

Datadog is a monitoring and analysis platform designed for cloud applications. It is focused on providing services to companies such as monitoring applications, systems, infrastructure, logs, and security (DCFmodeling, 2025). Datadog also provide full monitoring functions, real-time analysis performance, customized dashboards, an alarm system, and log management. Datadog was founded by Wireless Generation, an American company, by Olivier Pomel and Alexis. The Headquarters of Datadog is in New York, USA. Initially, Datadog focused on providing cloud infrastructure monitoring services, offering dashboards, alerts, and metric visualizations (News, 2025). Therefore, Datadog provides services to various companies, including Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform, Red Hat OpenShift, VMware and OpenStack. In 2016, Datadog developed a new technique, Application Performance Monitoring, the first full-stack monitoring solution. Until now, Datadog has adopted various technologies to keep upgrading and improving its system. For example, tie analytics to data monitoring, an Artificial Intelligence-based application testing platform, an application security platform, an API observability platform, and AI-powered data observability.

2.2.2.1 Features

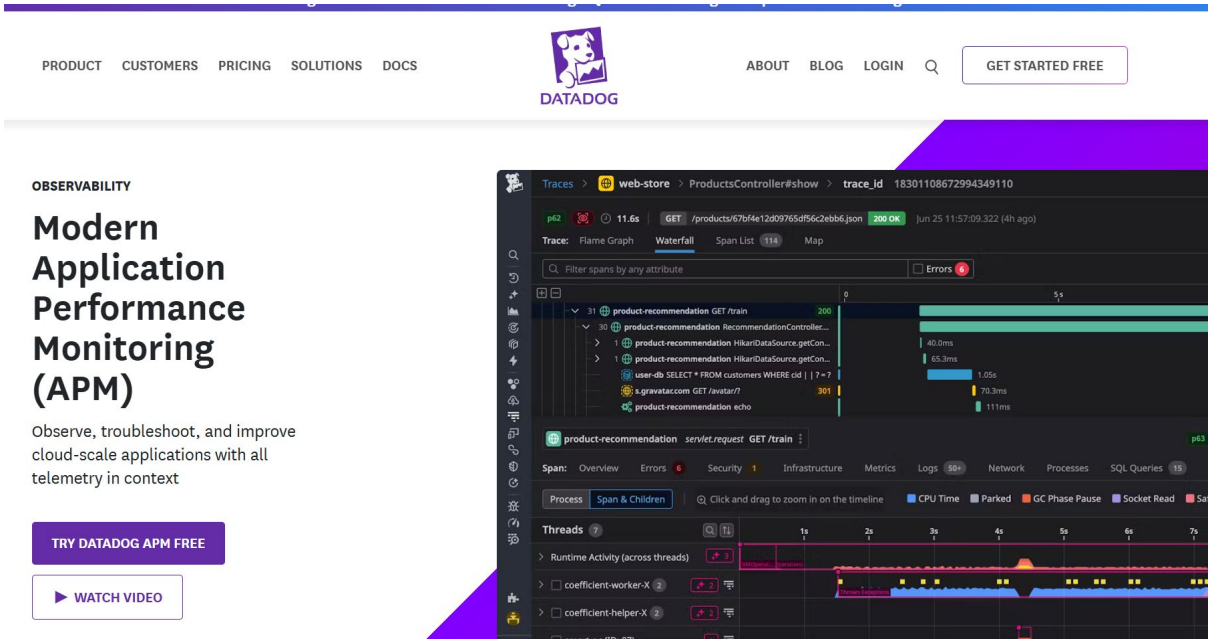


Figure 17: Application Performance Monitoring of Datadog's Features (datadog.com)

Datadog is a successful monitoring platform because it is a leading monitoring technology with comprehensive analytical capabilities, such as infrastructure monitoring that extends to application performance monitoring (APM). This function includes the network, containers, serverless architecture, cloud cost management, database monitoring, analysis code performance and performance analysis. Next, Datadog also has functions for log collection, which include log management capabilities. This function enables users to reduce the troubleshooting and analytics times by using search, filter, analyze, and process. Datadog provides a control panel that has process, enrich, and route all logs functions for users. This means users can customize the reference data to add more context and make it more actionable to build a consistent, structured dataset.

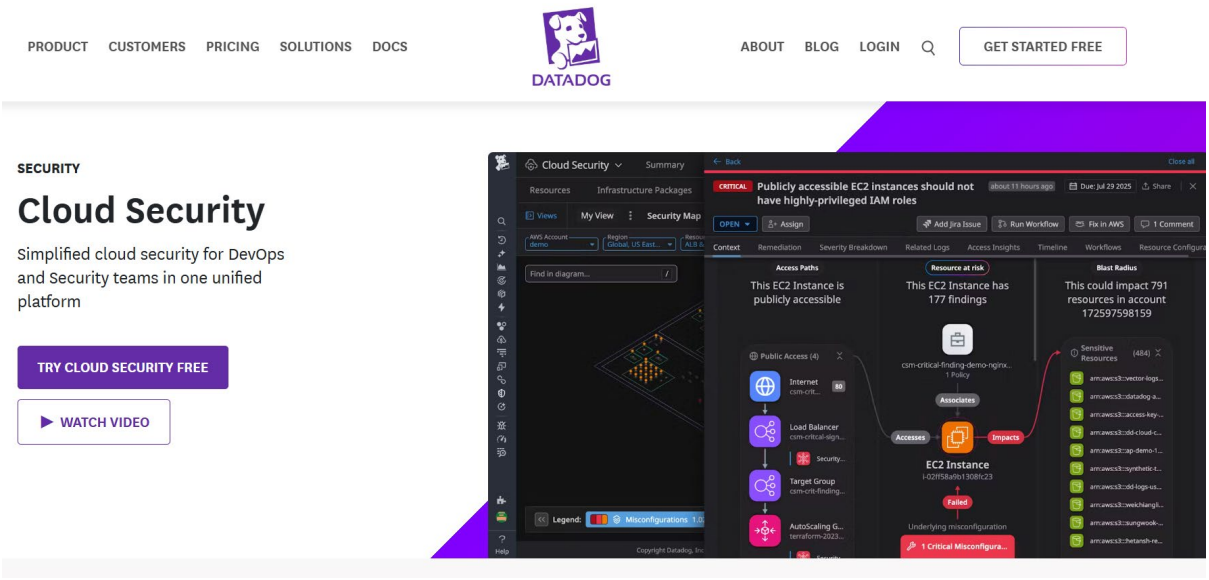


Figure 18: Cloud Security Posture Management (CSPM) of Datadog (datadog.com)

In the security area of networks, Datadog has provided a security protection solution, which is cloud security posture management (CSPM). The main function for cloud security posture management (CSPM) is using agentless technology within the cloud to detect all infrastructure for vulnerabilities, misconfigurations, identity risks and compliance violations. Not only that, but it also provides a security protection function that secures the cloud infrastructure from IAM-based attacks. For example, verify a full detection identity risk, including resource access, resource permissions management, and remediate for efficient mitigation. In the latest version of Datadog, it has integrated artificial intelligence (AI) into the detection system to achieve automated and intelligent operation functions, such as Bit AI Agents and Bit AI SRE. Bit Ai is an intelligent AI component that can be used for investigating alerts, suggestions provided, detecting security signals, and operational awareness. The example functions of Bit Ai are auto classification of alarm levels, auto detection of error signals, auto provide suggestions to fix errors, identify the potential risks, and auto analyze the detection result.

2.2.2.2 Strength and Weakness

Strength

The strength of Datadog is a full platform that integrates cloud native architecture, powerful integration, AI, Telligent analytics and automation. Datadog provides a unified platform strategy that integrates infrastructure monitoring, application performance monitoring (APM), log management, user experience monitoring (RUM), and network performance monitoring (NPM) (Scott Charles, 2025). This advantage can help users reduce the time to perform a suitable solution function without switching between multiple systems. In addition, the AI function helps Datadog collect and show the data results without manual settings (GuruFocus Research, 2024). These functions can help users move from a passive response to a proactive response before the issue affects the business.

Weakness

The weakness of Datadog is the high-cost requirement and a highly technical team. Datadog is one of the most expensive website monitoring systems in the market (SWOTAnalysis.com, 2025). The price of Datadog is based on the number of monitored hosts, the volume of tracking, and the number of events. The cost issues are the primary problem that most companies will focus on; if the cost is over the company's budget, there is a risk of budget overruns. Next, the high complexity component and functions in Datadog require a high technical knowledge and skill requirement for basic use of Datadog's functions (Scott Charles, 2025). An experienced and technically skilled team will increase the company's need for significant investment and personnel training in website monitoring projects. For a non-technical background, the team will have a big challenge in understanding the meaning of configuration options, dashboard components, and technical jargon (GuruFocus Research, 2024).

2.2.3 UptimeRobot

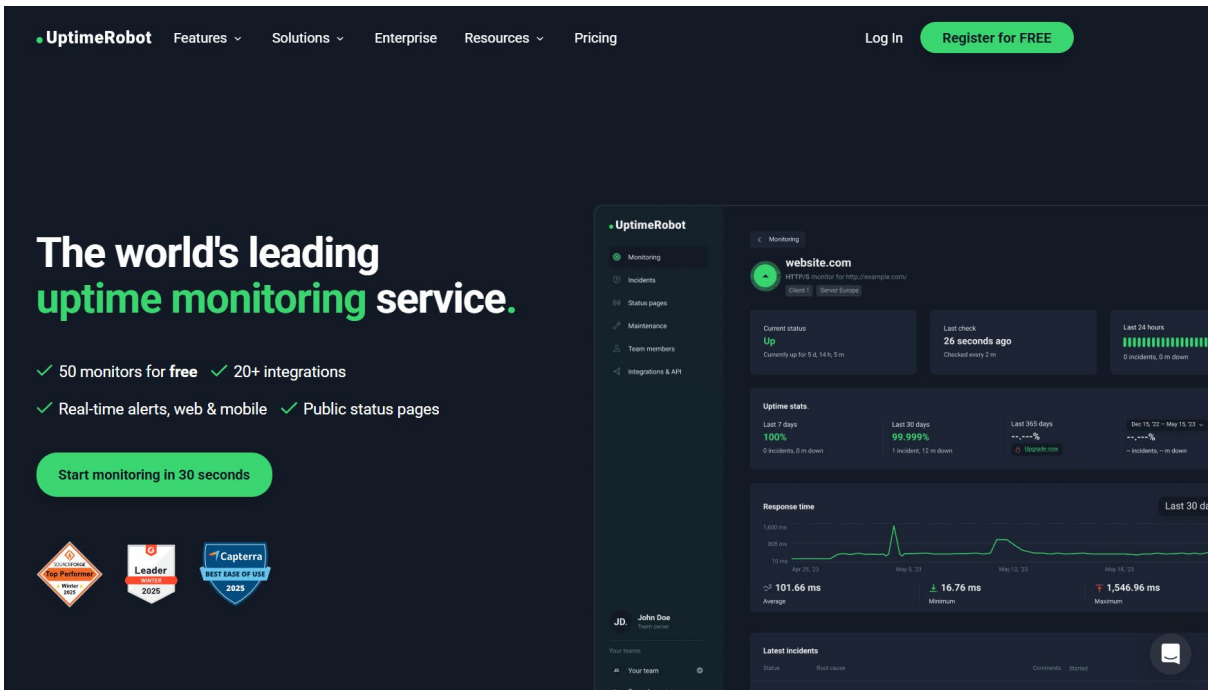


Figure 19: UptimeRobot Official Website (uptimerobot.com)

UptimeRobot is a highly cost-effective website monitoring tool that provides uptime monitoring and alert services. It can provide various types of websites monitoring checks, including HTTPS, ping, port SSL, and keyword monitoring (Chowdhary, 2025). UptimeRobot was founded in 2010 in Istanbul, Turkey, by Daniel Rimille and Umut Muhaddisoglu. Initially, the organization's main goal was to ensure that website monitoring services continued to operate even if the target website experienced downtime. In the beginning, UptimeRobot provided a fully comprehensive detection of website functions for users, such as identifying the website's errors by using keywords, and an alert system. It has become one of the most popular website monitoring tools globally. Until 2019, it was purchased by Itrinity, which is a Czech conglomerate company, and UptimeRobot needed to face some challenges, such as the increasing DDoS attacks, limitations of quantity monitoring, and network downtime issues (*Website Monitoring*, 2025). Nowadays, UptimeRobot has become an advanced global monitoring service from a normal, simple website detection tool.

2.2.3.1 Features

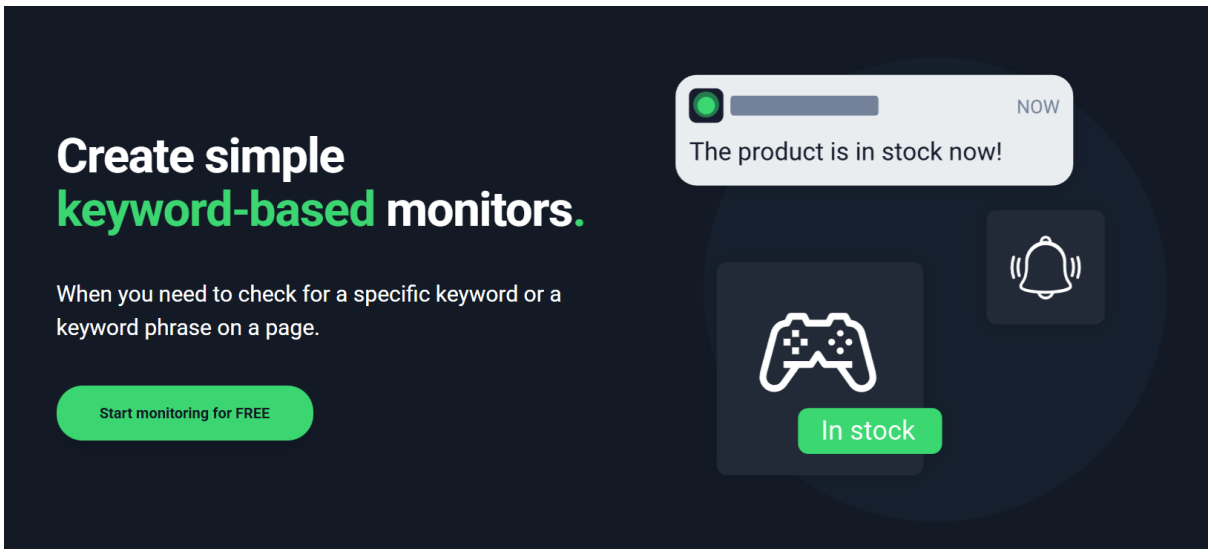


Figure 20: Keyword Monitor of UptimeRobot (uptimerobot.com)

UptimeRobot has advanced monitoring technology, so it has become one of the most popular website monitoring services in the world. For example, the advanced monitoring technology includes keyword monitoring, SSL monitoring, Ping monitoring, Port monitoring, and Domain monitoring. For the keyword monitoring service, users can be based on the specific keyword to collect the presence or absence of any phrase, such as set up an unlimited number of monitors according whether the keyword is or is not included in the page HTML code (Uptime Robot, 2025). By following the user's likes, UptimeRobot is able to let users set up customised keyword monitoring. Not only that, UptimeRobot enables users to monitor the SSL Certificate to ensure the securely encrypted website is secure, such as without the "https" in front of the URL, the website is purposely dangerous (HuntScreens, 2025). It can provide an automated SSL monitoring for the target website to identify the website's SSL certificate expirations, SSL certificate errors, and maintain seamless website accessibility and security.

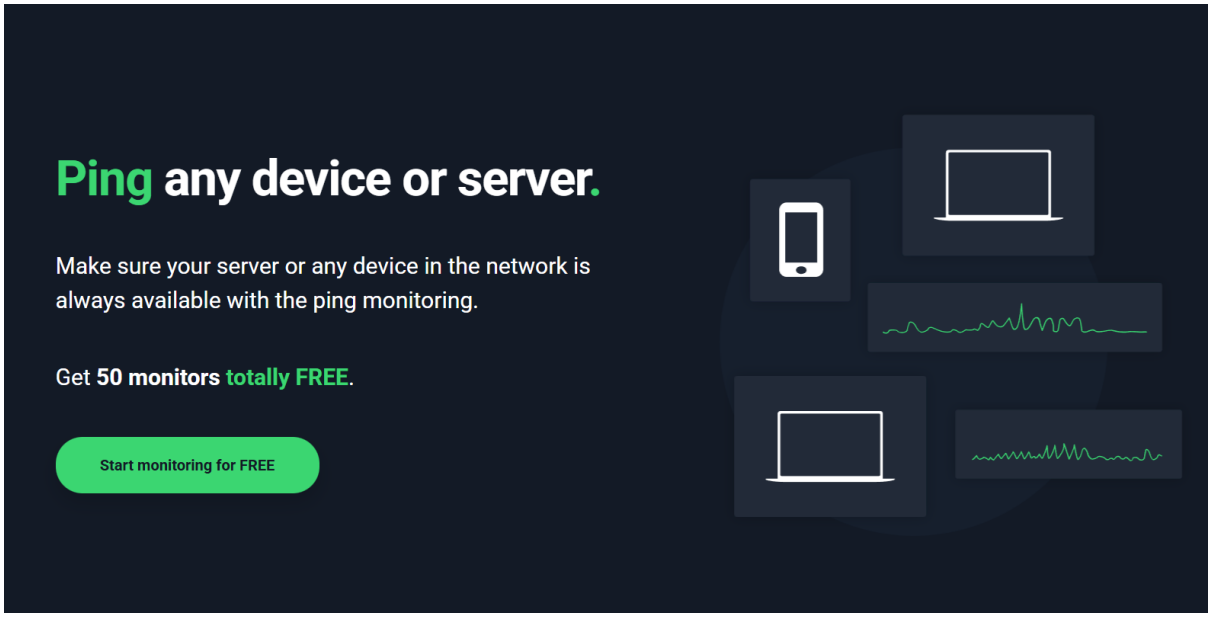


Figure 21: Ping Monitor of UptimeRobot (uptimerobot.com)

In addition, the Ping Monitor function of UptimeRobot can help ensure the availability of the server or devices with the ping monitor. When the devices are connected to a reliable network and the server's response time, the ping monitor can help ensure the safety and availability of the website. Furthermore, Port Monitor is one of the functions in the UptimeRobot. It can monitor any ports and make sure the email service runs smoothly on the server. For example, the type of ports, HTTPS (443), SMTP (25), FTP (21), SSH (22), DNS (23), IMAP (143), and custom. Lastly, the domain monitor can help ensure the online presence is secure and avoid any potential damage by monitoring the domain's expiration date.

2.2.3.2 Strength and Weakness

Strength

The strength of UptimeRobot is cost-effectiveness, multi-alert options, and global monitoring infrastructure. UptimeRobot can provide a flexible notification ecosystem that provides an extensive array of alert methods, such as SMS messages, email sending, team collaboration on Slack, Microsoft Teams, and message applications like Telegram. High flexibility on the alert system can ensure the technical team can receive the important website downtime notification in an urgent situation and reduce the response times to able the effects of the website downtime. Not only that, UptimeRobot is using over 30 monitoring stations worldwide to enable a comprehensive assessment of a website's performance from different areas' perspectives. It can increase the accuracy of the website detection from the global footprint and the latency problems of Asian or European users.

Weakness

The weakness of UptimeRobot is limited functions on the advanced website monitoring, and limited generation of basic detection reports. UptimeRobot lacks the ability to provide some important advanced website monitoring functions, including synthetic transaction monitoring. In addition, the detection result report from UptimeRobot is simple and does not provide any detailed performance metrics, such as the page load time of the target website, refresh times, and geographic performance variations. This weakness will pose a big challenge for users who need to make a deeper analysis and a deeper decision. Furthermore, UptimeRobot does not provide any integration of artificial intelligence or machine learning functions; it is performed based on the rule-based alerting mechanisms. So, UptimeRobot will perform any automation analysis on the alert message and alert patterns, which will need some manual operations for a meaningful understanding of monitoring data.

2.2.4 Similar System Comparison Table

Table 2: Comparison of Similar System

Aspect	Pingdom	Datadog	UptimeRobot
Type	Tool of website monitoring and performance optimization	Cloud Monitoring	Website Monitoring Service
Primary Purpose	Provide performance, uptime checking of the website	Provide full information of the cloud application, logs, and security	Ping Monitor, Port Monitor, Domain Monitor, Uptime Monitor
Interaction Level	Medium (Provide both synthetic monitoring and real user monitoring)	High (Provide full relatively comprehensive capabilities of website monitoring)	Low (basic detection and simple UI)
Deployment Method	Cloud-based service for detecting and monitoring globally	Cloud-based platform	Cloud-based service
Attack Detection Focus	Focus on the availability and performance of the website	Comprehensive	Focus detection on the basic attack

Log Visualization	Provide page load times and performance metrics of detailed reports	Provide a customized dashboard and log management reports	Downtime of the website reports
Alert System	Email and up to 4 notifications of SMS messages	Alert system with AI notification (automation classification, automation analysis)	Email, SMS, Telegram, Push, and Webhook
Target Audience	Large enterprise	High skill and professional teams	Small business

The summary of the similar system comparison table above, Pingdom is a tool for website monitoring and performance optimization. The goal of Pingdom is to provide a performance and uptime check of the website to the target users. It has combined synthetic monitoring and real user monitoring, resulting in a medium interaction level. The deployment method of Pingdom is a cloud-based service for detecting and monitoring globally. It can provide the availability and performance of the website to the target user. When Pingdom detects abnormal data and performance of the target website, it will use email and SMS messages to notify the user for troubleshooting, but the SMS messages will only include up to 4 notifications. The suitable target audience of Pingdom is for businesses that more on website performance and status.

Datadog is a type of cloud monitoring that focuses on detecting full information about the cloud application, logs, and security. It can provide full, relatively comprehensive capabilities of website monitoring, resulting in a high interaction level. Datadog is using a cloud-based platform to provide a full, comprehensive attack detection for users. User can base their log management report on a log visualization. When Datadog detects abnormal data and performance of the target website, it will provide an AI-powered notification to implement automation classification and automation analysis functions for users. Therefore, Datadog need

highly skilled and professional teams as a target audience because its integrated and advanced monitoring capabilities have become a challenge for non-technical teams.

UptimeRobot is a website monitoring service that focuses on ping monitor, port monitor, domain monitor, and uptime monitor. It can only complete a basic detection for a website's security and simple interface, with the interaction level being low. The deployment method of UptimeRobot is a cloud-based service, and the primary attack detection will be on the basic attack without any complex availability checks. It will record the uptime and downtime of the website and identify reports to complete the log visualization. Email, SMS, Telegram, Push, and Webhook are the methods that UptimeRobot can transmit the website information to the alert system. Finally, the target audience is small businesses, including the developer and the individual.

2.3 Technical Research

2.3.1 Hardware and Software Requirement

Hardware Requirement

Table 3: Hardware Requirements

Component	Specification
Processor	Intel Core i5 or equivalent
RAM	8 GB
Storage	At least 256 GB
GPU	Optional NVIDIA GTX 1050 or equivalent

The goal for hardware requirements is to ensure that the website monitoring functions, and machine learning algorithms can perform smoothly, so that it will focus more on the balance between performance and efficiency. The hardware requirement of the processor is Intel Core i5 or equivalent to provide a sufficient environment to run programs as a developer. For example, run back-end APIs, database, perform website monitoring functions and process user request tasks. A minimum of 8 GB of RAM can provide multitasking and dataset management capabilities during the development and testing phases.

Having up to 256 GB of storage on an SSD can provide enough space for developers to store program files, packages and logs and increase the data access speeds, so that it will reduce the data access waiting time. Next, for the hardware requirements of the GPU, a NVIDIA GTX 1050 GPU or equivalent can provide high-speed model training processes and advanced algorithms, such as machine learning algorithms. This configuration, which offers a balance between performance and efficiency, is well-suited for the scale and complexity of real-time website monitoring and alerting system projects.

Software Requirements

Table 4: Software Requirements

Component	Specification
Programming Language	Python and JavaScript
Frameworks	Flask (Backend), React.js (Frontend)
IDE	Visual Studio Code
Database	SQLite
Operating System	Windows 11 Home Single Language
Web Server	Flask built in development server
Web Browser	Google Chrome

The software requirement can provide a level of development for MonWeb's real-time website monitoring and alert system to ensure a certain level of functionality, ease of use and compatibility. Python has been selected to become a backend programming language for this project because it can provide a full library support for network monitoring, data processing, and artificial intelligence, which enables implementing HTTPS(S) and ICMP detection functions. Next, the front-end development will integrate JavaScript and React.js to build a highly interactive web interface and user-friendly UI as a primary goal. JavaScript can implement an interactive user interface and provide advanced Web techniques for developers. Reach.js can implement a dynamic and responsive user interface that can visually present website uptime, response time, and alert notifications in real time as a front-end development. Not only that, but Flask will also be adopted as the backend framework because it can build a handle website monitoring requests and enable lightweight, flexible, and efficient development for developers.

The integrated development environment (IDE) that will be used in this project is Visual Studio Code (VS Code) because it can provide full extensions, particularly developed for Python and JavaScript programming, thus it can help with debugging, ensure the development process, and project management. For the data storage, SQLite is selected as the database system for this project, which has simplicity, a configuration nature, and suitability. SQLite is suitable for development phases of data storage, such as log files and historical data, during development. The operating system that has been chosen for this project is Windows 11 Home Single Language. This version can support the selected frameworks, development tools and browsers, such as Google Chrome, Firefox, and Edge.

The web server of Flask that has been chosen for this project, its internal development server, can help to test the API endpoints and integrate back-end and front-end components to ensure sufficient performance for prototyping and functional testing is suitable. Lastly, Google Chrome is chosen as the web browser for this project for testing and debugging the interface. Google Chrome can provide a full set of reliable developer tools for debugging front-end and web requests. The goal of the software requirement for this project is to achieve the project goals efficiently, balancing functionality and scalability.

2.3.2 Programming Language and Framework



Figure 22: Logo of Python (greenpegacademiy.com)

Python and JavaScript are the programming languages and frameworks that have been selected in this project. Flash will be used as a back-end framework, and React.js will be used as a front-end framework in this project. Python has been selected as the programming language and framework in this project because it can provide a suitable ecosystem of libraries and frameworks for data processing and artificial intelligence for a website monitoring system (Mahalaxmi et al., 2023). For example, Python can supply a standard library that can implement the website monitoring and analysis and natural language processing components for this project, such as Natural Language Toolkit (NLTK), request libraries for network monitoring, urllib3 libraries for HTTP(S) requests, PyTorch, and scikit-learn (Singh & Dhir, 2019). In addition, Python can also provide libraries for data processing, which can process and analyze the data generated by website monitoring and detect patterns to generate meaningful insights.



Figure 23: Logo of JavaScript (JS) (codup.co)

JavaScript has been selected as the programming language and framework in this project because it can build dynamic, interactive user interfaces. For example, JavaScript can provide an event-driven architecture to perform a real-time update of the website monitoring data's result, showing the target website's status changes, perform alert visualization, and display the performance of the target website (Gopal et al., 2022). Not only that, but JavaScript also has a comprehensive ecosystem of libraries and tools for data visualization to show the website monitoring data, website response times, and system performance metrics, such as Chart.js, D3.js, and Highcharts (Smilkov et al., 2019). The combination of React.js and JavaScript can enhance the user experience when using the website monitoring system. React.js can provide a component-based architecture to allow users to create reusable UI components, thereby building a powerful monitoring dashboard with multiple elements. Next, React.js has an insightful community and comprehensive documentation for users to access support resources, tutorials, and third-party components, thereby users can reduce the troubleshooting times when facing challenges in building a monitoring dashboard.



Figure 24: Logo of Flask (fiverr.com)

The backend framework of Flask is lightweight and flexible, which is one of the considerations for selection in this project. Flask is an imposing micro framework with unnecessary constraints or dependencies, which allows web development to be more flexible in designing the system architecture and based on the specific project requirements, to make changes (Tewari, 2025). Afterwards, Flask can integrate with Python's artificial intelligence functions and data processing libraries to implement an AI-driven monitoring analysis and NLP components without complexity (Dhanunjay Bhavani et al., 2024). Therefore, Python and Flask were used as the backend frameworks and JavaScript and React.js as the frontend frameworks in this project. The combination of these two can meet all the technical requirements of the system.

2.3.3 IDE Interactive Development Environment



Figure 25: Logo of Visual Studio Code (VS Code) (medium.com)

The Integrated Development Environment (IDE) selected for this project is Visual Studio Code (VS Code). Visual Studio Code is a versatile, cross-platform compatible and robust ecosystem, as it can support a large volume of programming languages, including the programming languages that are needed for this project, which are Python and JavaScript (Tan et al., 2023). Visual Studio Code's low resource consumption and high performance make it suitable for development on various systems (Johnson, 2019). Furthermore, its comprehensive ecosystem of extensions is more aligned with the integrated development environment required for this project.

The key extensions are also one of the major advantages of Visual Studio Code, as they can effectively improve the efficiency of developing MonWeb projects. For example, the

unified development environment of Visual Studio Code can support users without eliminating context switching between tools to reduce development times because VS Code is able to support Python and JavaScript as a backend and frontend (Durelli et al., 2022). Furthermore, VS Code has a key extension for developing optimization, artificial intelligence, and natural language processing, which is IntelliSense that is provided by Microsoft (AlanRynne, 2025). IntelliSense provides comprehensive features to support users in enhancing the quality of the development code and reducing the development time, such as advanced intelligent code completion, syntax highlighting, automatic formatting, and error reporting. The key extensions can significantly improve efficiency and optimize code quality during the development phase of this project.

2.3.4 Database Management System



Figure 26: Logo of SQLite (medium.com)

SQLite is a lightweight, serverless database management system, primarily suitable for small projects. It is also one of the most widely used databases in the world because it can be built into various devices and programs, such as mobile devices, desktop applications and web browsers (Gaffney et al., 2022). Therefore, SQLite is the most suitable database management system for the website monitoring alert system in this project.

One of the advantages of using SQLite as a database management system is its serverless architecture. The database can be stored in a single disk file because it is able to do so without a separate server process or operating system, and it helps to reduce the network

overhead (Bhosale & Patil, 2015). Not only that, but it is not a requirement for management and configuration files, which can help developers reduce time on configuration or installation. The developer will focus on code development phases by using the extra time from configuration.

Another advantage that SQLite has is that it provides high-performance database management. It can provide an optimized solution for low memory environments, leading to faster database operations of the file system input and output. Next, developers are also able to switch on multiple operating systems for using the SQLite database management system because the data stored in the dataset is stored in a single file, which allows for Cross-Platform Compatibility (Purohith et al., n.d.). Lastly, SQLite as a database management system in this project is suitable. In the future, as the database requirements of projects increase with development, it can also greatly help developers easily migrate to more powerful database systems.

2.3.5 Operating System



Figure 27: Logo of Windows 11 (wonderfulengineering.com)

Windows 11 Home Single Language is the primary option for the operating system in this project because it is a modern and fully functional operating system developed by Windows, providing a stable and lightweight development environment for a website monitoring alert system (Wallace, 2020). Its minimal resource consumption can improve the performance and efficiency of website monitoring systems in detection, as the operating

system's limitation to a single display language reduces the burden of language packs on the system.

The advantage of selecting Windows 11 Home Single Language as the operating system for this project is the high compatibility with development tools, such as Python, JavaScript, Flask, React.js, Visual Studio Code and SQLite. It can fully support the programming language and framework needed for developing AI, NLP, and website monitoring functions of libraries. Next, it can provide a safe development environment and internal safety identity system, such as TPM 2.0, Secure Boot, and Windows Defender, which are used to prevent threats that may compromise the website monitoring alert system. Overall, Windows 11 Home Single Language is a suitable operating system for this project and ensures a stable, secure, and efficient operating system for a development environment.

2.3.6 Web Server

Flask is the web server that has been selected in this project. In the development phases, Flask can provide a lightweight and easy-to-configure environment to enable testing the functions of the website monitoring alert system (Kgil & Mudge, n.d.). When developers make an upgrade for the website monitoring alert system, the web server will auto-refresh and update, thereby enhancing the development process and efficiency. The goal of Flask is to support developers in building an API endpoint and to easily integrate with front ends such as React.js.

For production environment deployments, a robust web server is essential, requiring careful selection, including Gunicorn and Nginx. These servers offer superior performance, stability, and security, and can handle multiple processes simultaneously, highlighting their importance in managing concurrent user requests, website monitoring requests, and alerts. The combination of these servers ensures high availability and scalability for the real-time monitoring system. Flask, as the web server architecture, provides a solid foundation for the MonWeb website monitoring system, supporting its core functions of real-time website monitoring, alert generation, and user-friendly reporting, while ensuring performance, security, and scalability requirements are met.

2.3.7 Web Browser

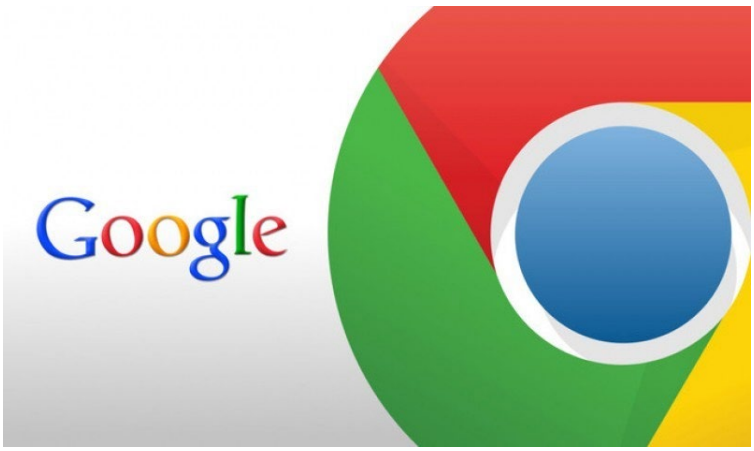


Figure 28: Logo of Google Chrome (techspot.com)

Google Chrome is the browser that was chosen for this project for testing and interacting with the system's front-end interface. Google Chrome is a popular browser with comprehensive developer tools for debugging web applications, analyzing network requests, monitoring website performance, and inspecting website elements throughout this project. These tools play a vital role in both the development and testing phases. For example, the developer tools are the DOM inspector, network activity monitor and JavaScript debugger.

Next, Google Chrome also provide a JavaScript engine to perform JavaScript code, thereby it can enhance the performance and show the real-time monitoring data, alerts, and interactive visualizations. Google Chrome also support using key extensions for extra testing and programs to increase productivity and efficiency. Therefore, Google Chrome is an essential tool for developing website monitoring systems.

CHAPTER 3: METHODOLOGY

3.1 System Development Methodology

3.1.1 Agile Methodology



Figure 29: Agile Methodology (tristatetechnology.com)

Agile Methodology is one of the Software Development Methodologies that can be designed to enhance the development, continuous improvement, and rapid delivery of functional components (Laoyan, 2025). Compared with the traditional methods that need to complete each phase sequentially, agile methodology can provide adaptive planning and incremental development. In 1957, people needed to build computer programs by continually improving the development process, thereby providing an interactive and incremental method (GeeksforGeeks, 2023). Next, people are trying to use adaptive software development and evolutionary project management to enhance and find a method to improve software development. In 1990, agile methodology was published by 17 software developers who could replace the more linear product development process.

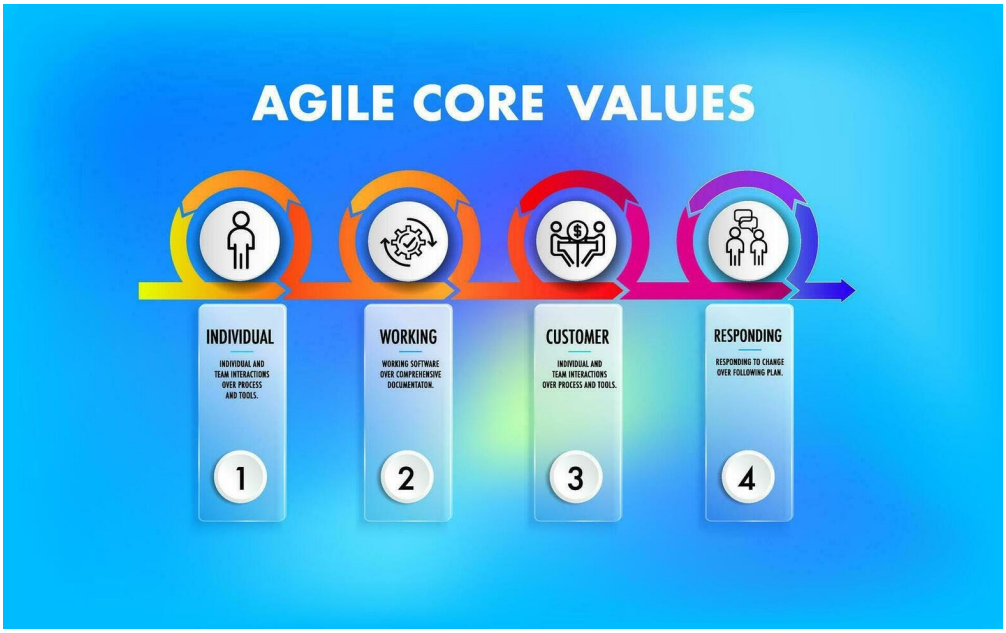


Figure 30: Value of Agile Methodology (vecteezy.com)

The target used of agile methodology for a project that involves dynamic requirements, fast-changing environments, and keeping the project system. There are a few main values of agile project management that can affect the development teams to collaborate, deliver software, and adapt to evolving requirements. The first value of agile project management is individuals over processes and tools. The goal of this principle is the importance of communication, teamwork, and human collaboration in performing the project's tasks. A technical team with strong creativity, decision-making ability, and adaptability can help improve project development activities (Laoyan, 2025). At the same time, technical team members can share their ideas to achieve close cooperation and jointly solve difficulties, rather than constantly following a fixed development process.

The second value of agile project management is working software over comprehensive documentation. Compared with the traditional development models, the documentation record will take most of the project time, and it will affect the actual system development planning (GeeksforGeeks, 2023). So, the aim of agile methodology is to balance the time usage for documentation, thereby it can submit increments of working software that can be tested, reviewed, and improved quickly. The next value of agile project management is customer collaboration over contract negotiation. In an agile methodology, close collaboration with clients becomes more important throughout the entire project lifecycle, rather than solely

through pre-planned contractual requirements. This approach of maintaining contact with clients enhances their role in guiding the software's development and allows for continuous adjustments based on their preferences and evolving expectations, ensuring that the project is not bound by rigid contracts (Makwana, 2025).

The last value of agile project management is responding to change by following a plan. The importance of responding to change is that it ensures the project system can always adapt to market conditions and continuously adopt effective strategies to keep up with market changes, even in the last stages of development (Laoyan, 2025). One of the benefits of agile project management is that it provides team flexibility, allowing teams to quickly adjust strategies and workflows in response to changes without derailing the entire project.

3.1.2 Waterfall Methodology

Waterfall Methodology is one of the Software Development Methodologies that can be used for software developers to design, develop, and test as a process to improve the software project (Atlassian, 2025). Another meaning of waterfall methodology is traditional methods for software development; it is one of the popular classic methods of the software development lifecycle in the software engineering area. The Waterfall Methodology originated from a paper on software engineering proposed by Winston Royce in 1970. In his paper, Royce demonstrated that the Waterfall Methodology describes the phased approach to software development, presenting the specific steps of each phase in a flowing "waterfall" format. The example of each process phase for software development is analysis, design, implementation, testing and maintenance (Nicolai, 2023).

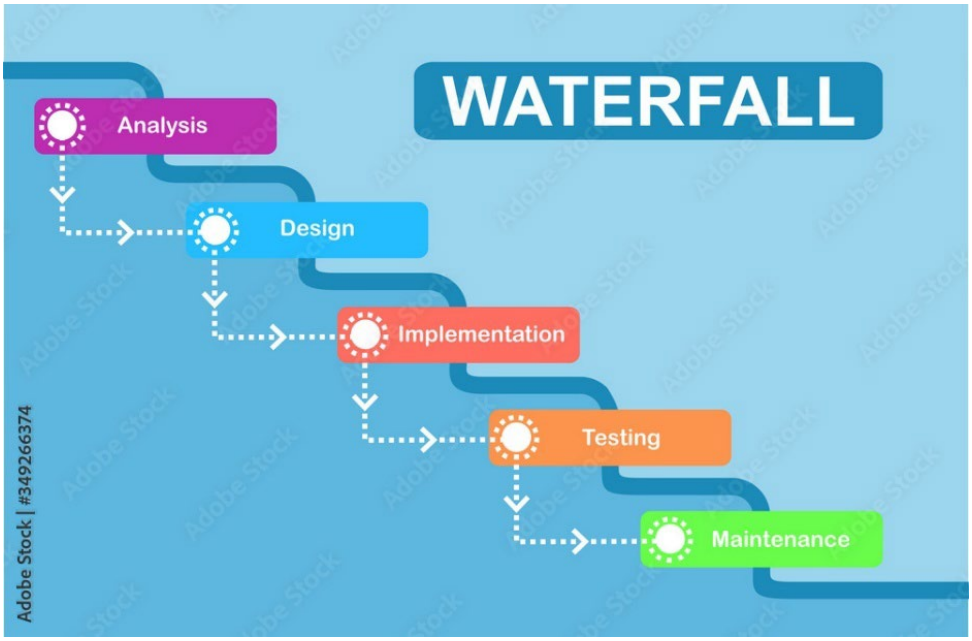


Figure 31: Process Phases of Waterfall Methodology (stock.adobe.com)

The process phases of the Waterfall Methodology are to provide a progression that flows steadily downward through distinct phases for developers. Therefore, each process phase must be fully completed before access to the next phase, its method can help to ensure the emphasised planning, documentation, and rigid structure (Nicolai, 2023). The first phase of the Waterfall Methodology is analysis. This phase is used to collect and record the functional requirements of the project. Not only that, but it can also help to determine the target audience, target system specification, and confirm the planning for the project. The second phase of the Waterfall Methodology is design. These phases can help developers to prepare and design the system architecture, such as system flow, database schemas, UI design, API endpoints and data flow (Kirvan et al., 2024).

In addition, the third phase of the Waterfall Methodology is implementation. The primary task for this phase is to develop a code based on the process phase 2, which is the design specifications. In this implementation phase, it is necessary to integrate the third-party tools and perform unit testing for components (GeeksforGeeks, 2023). The next phase of the Waterfall Methodology is testing. Developers need to perform the integration of the system and user acceptance testing (UAT), thereby fixing the system and improving its performance. The last process phases of the Waterfall Methodology are deployment and maintenance. It will need to train the first users for creating a full user guide, record the installation system

steps, and prepare for IT admins. Meanwhile, developers need to collect feedback from the first users, which can help enhance the system features based on the feedback (Rory, 2022). Developers also need to prepare for maintenance and updates when users face any bug or error.

3.1.3 Feature Driven Development (FDD)

Feature-driven development is one of the types of software development methodology, which is a lightweight or agile method for developing method and focus on the customer requirements (Team, 2021). In 1997, Jeff De Luca published a report on the need for software development projects that could meet the needs of large organizations and accommodate 50 people. Feature-driven development (FDD) ultimately became the preferred and popular methodology for managing responsible software projects. FDD employs an iterative, incremental development approach to achieve agile and flexible projects. The core principle of FDD is "build by features," which allows the entire development process to be based on the identification, design, and implementation of various features that meet the actual value requirements of the client (Harness, 2025).

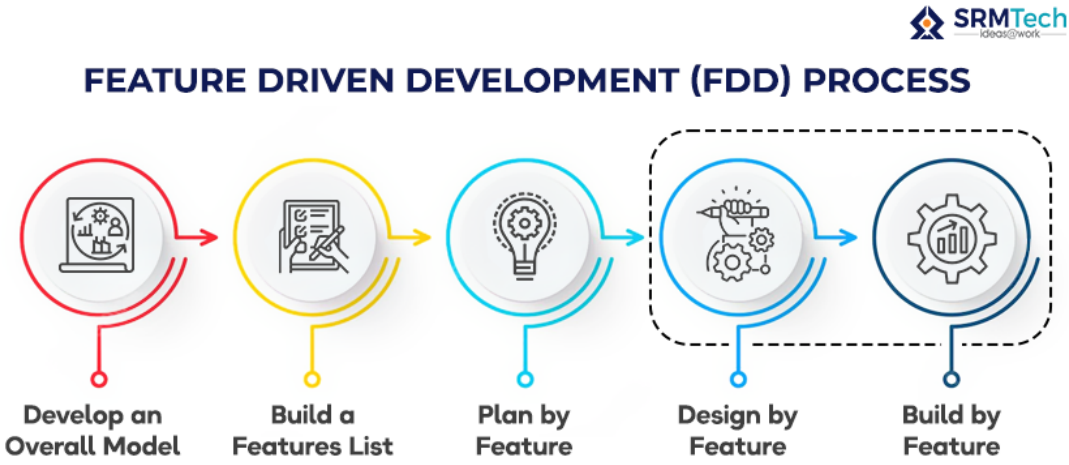


Figure 32: Process of Feature Driven Development (FDD) (srmtech.com)

There are a few main sequential processes that are used to guide the software development project. The first process of Feature-driven development (FDD) is to develop an overall model. This phase is based on the project requirements, and then developers and domain experts collaborate to build a high-quality system model to meet the requirements

(International Agile Federation, 2024). The developer team also need to build an overall architecture to identify the key components and guide the subsequent feature-based planning. The second process of Feature-driven development (FDD) is building a features list. In this phase, the developer team will need to create a feature list that is based on the overall model and project requirements. The functions list will be separated into small, manageable client-valued features, which can help the developer team organise, prioritise, and estimate development tasks (CodeLucky, 2025).

The third process of Feature-driven development (FDD) is plan by feature. In this phase, the project manager will base their assignment on the skill sets, complexity, and dependencies of developers to assign tasks. The goal of this assignment is to create an accurate development plan, such as the resource allocation, timelines, and sequencing of feature sets (admin, 2025). The next process of Feature-driven development (FDD) is design by feature. In this phase, developers will create a temporary feature small team, which is used to design specific features selected for development. The aim of the temporary feature team is to review the design constraints, establish component interactions, and prepare detailed design packages. Lastly, the last process of Feature-driven development (FDD) is built by feature. Every feature requirement will need to be implemented, tested, and reviewed. A developer can perform unit testing to ensure the functions of the project and reduce the number of problems (Harness, 2025). This iterative cycle will repeat for each of the features and add to the growing system, thereby enabling continuous delivery and early visibility of progress.

3.1.4 Methodology Comparison

Table 5: Comparison of Methodology

Aspect	Agile	Waterfall	Feature Driven Development (FDD)
Approach	Adaptive planning of iterative and incremental development	Linear sequential complete distinct phases	Focus on customer requirements to perform iterative and incremental development
Focus	Constant adjustments and changes, suggestions from clients	Follow the established plan	Targeting the actual value of customers
Development Time	Quick Feature Component Completion Time	Complete each stage slowly and sequentially.	Delivery time depends on the complexity of the function.
Documentation	Priority task to complete the working software, reducing the need for documentation.	detailed collection of documents for each stage	Appropriate documentation and a focus on functional design
User Involvement	Maintain a high level of involvement throughout the entire development cycle	Low involvement until project requirements and testing	Focus on functional planning and outcome phases

Risk Management	In each iteration, perform risk management and mitigation	Poor risk management in the face of constant change	Based on the feature to perform risk management
Flexibility	High flexibility, capable of continuously adapting to future adjustment needs.	Low flexibility when each phase is completed	Medium Flexibility when facing change
Team Size	Small Team	Medium Team	Large Team
Complexity	Based on the iteration to complete the complex project	Clear requirement for complete project	Based on feature decomposition to manage complexity
Cost	Lower cost	High Cost	Based on feature resource allocation to decide the cost-effective
Scalability	Scalable with multiple teams	Scalable but not suitable for large project	Scalable for enterprise level projects
Suitability	Dynamic environments	Stable environments	Clear features and client-valued requirements with a large-scale project
Management	Collaborative and adaptive	Structure and plan driven	Feature owner driven

3.1.5 Chosen Methodology

The chosen methodology for the "MonWeb: a Real-Time Website Monitoring Alert System" is Agile Methodology. This is because Agile Methodology is more on iterative and incremental approach to software development, it has showed the flexibility, collaboration, and customer feedback. Therefore, Agile Methodology aligns well with project requirements and deliverables. Furthermore, Agile Methodology offers a relatively clear and defined development cycle for implementation, which are planning requirement, system design, develop into sprint cycles, testing and evaluation, deployment, review, and feedback. The each of the sprint will provide a functional module and improvement.

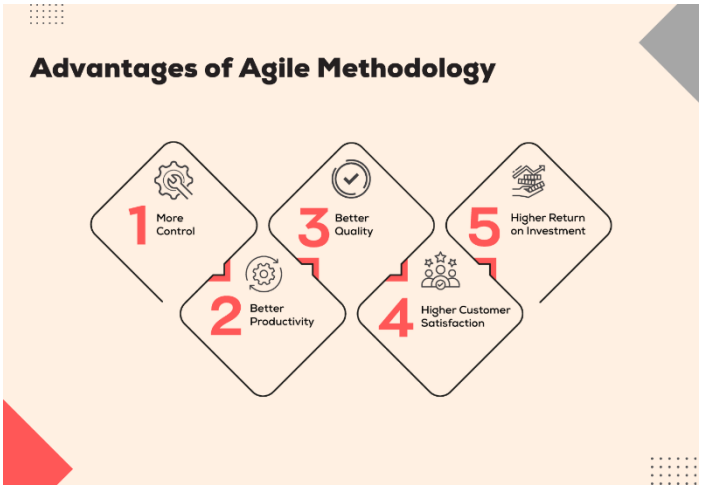


Figure 33: Advantages of Agile Methodology (inspirisys.com)

The first reason for selecting Agile Methodology is risk management and fixing the evolving project's requirements. In this project, the website monitoring and alerting system includes website request testing functionality, target website performance information detection functionality, alert issuance mechanism functionality, detection result analysis functionality, and integration of artificial intelligence (AI) and natural language processing (NLP) components. Agile's iterative nature enables developers to improve the weaknesses based on the testing results and user feedback. In addition, Agile's incremental approach can also help reduce the project's risk by identifying technical issues because of the technical complexity of real-time monitoring and artificial intelligence integration.

The benefit of using Agile Methodology as system development methodology in this project is adaptable on shift strategies without disrupting the flow of the project. Compare with the traditional software development methodology such as Waterfall Methodology, it is using a shifting strategy that is having a problem on disrupt the rest of the project planning. Therefore, it is one of the reasons that Agile Methodology become a popular method for software development methodology.

The next benefit is that Agile fosters collaborative teamwork in software development. Effective teamwork relies heavily on good communication, with face-to-face interaction being the best way to avoid the limitations of online communication that can hinder the acquisition of detailed information. While technology has made tremendous strides, a clear communication method is still essential for an efficient and collaborative technical team. For example, the technical team need an active communication to ensure they clearly understand the development task, progress updates, issues, and the method for solving the problem during the development phases, especially for multi-module systems.

The ultimate benefit of the Agile Methodology is that it prioritises customer needs. Software development projects originate from the requirements provided by clients to the technical team and the basic functionalities needed for a standard system. Therefore, continuous client attention and satisfaction are key motivators for the development team to complete the entire software development process. The Agile Methodology prioritises developing corresponding system functions quickly by focusing on customer needs. Therefore, the team also needs to continuously adapt to changes during this process. Furthermore, adjusting and testing software development is also a strategy to enhance functionality, quality, and performance.

3.2 Data Gathering Design

3.2.1 Survey

A survey is one of the data gathering methods by using a questionnaire, asking questions to a group of individuals (Maya, 2025). For example, a survey is a structured conversation that can ask the same question or different types of questions to multiple people, thereby it can provide an understanding of patterns, opinions, and behaviours. There are various media, such as pencil and paper, online forms, telephone surveys, and face-to-face interviews (Bhat, 2018). The aim of this survey is to collect the related data that helps gather valuable insight and feedback from multiple groups of people. Compared to various data collection methods, surveys are a less resource-intensive tool that can reach a broader target audience. Surveys can be tailored to the specific data collection requirements of a project, creating a series of relevant questions to understand the group's attitudes, behaviours, and characteristics towards the target project (Kate, 2025). Furthermore, surveys can help the team identify the project's main objectives, preventing them from losing momentum during software development.

Cover Page and Introduction

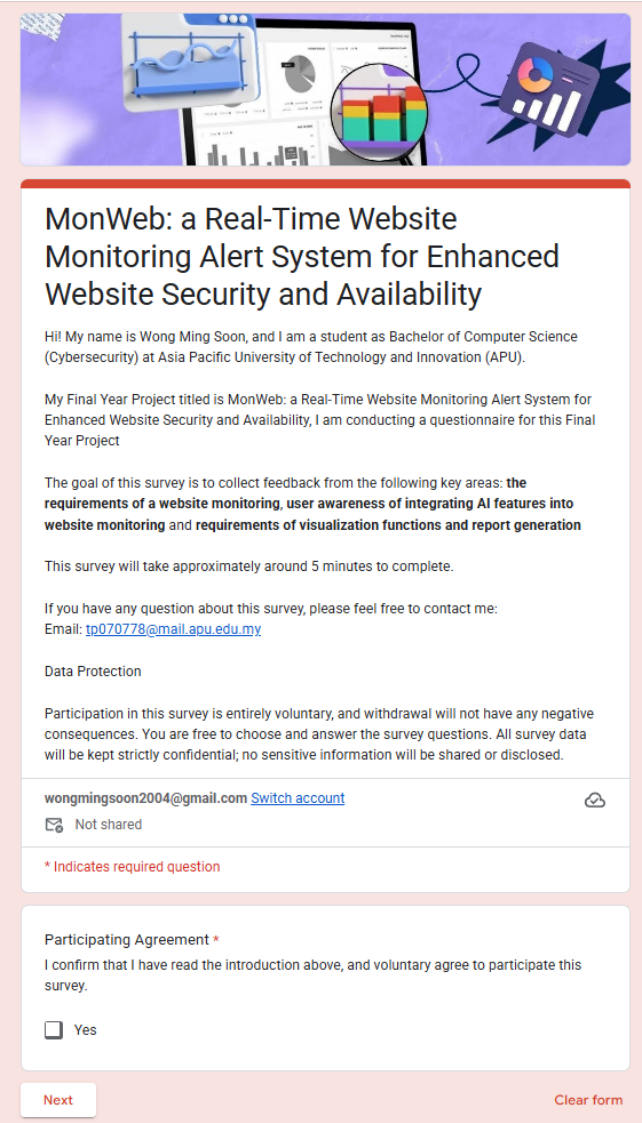


Figure 34: Cover Page and Introduction of Survey

The figure shows the introduction page of the survey that is designed for the development of MonWeb: a Real-Time Website Monitoring Alert System for Enhanced Website Security and Availability. The aim of this survey is to collect the feedback and patterns from target participants, including the website monitoring, AI integration, and visualisation.

The goal of this introduction page is to show a brief overview of the project and the objective of developing a real-time monitoring system that can increase website security and ensure website availability. The document also assures participants of data confidentiality,

stating that all data from this survey will not be leaked and will not have any negative impact on participants.

The page also emphasises that the survey is expected to take approximately 5 minutes to complete and clearly outlines its objectives, including gathering participants' understanding and needs regarding website monitoring systems, AI integration, and feedback on visualisation features. Contact information is also provided on this page to facilitate participants' inquiries, build trust in the survey, and ensure a friendly and professional page structure and tone.

Section 1: Participant Information

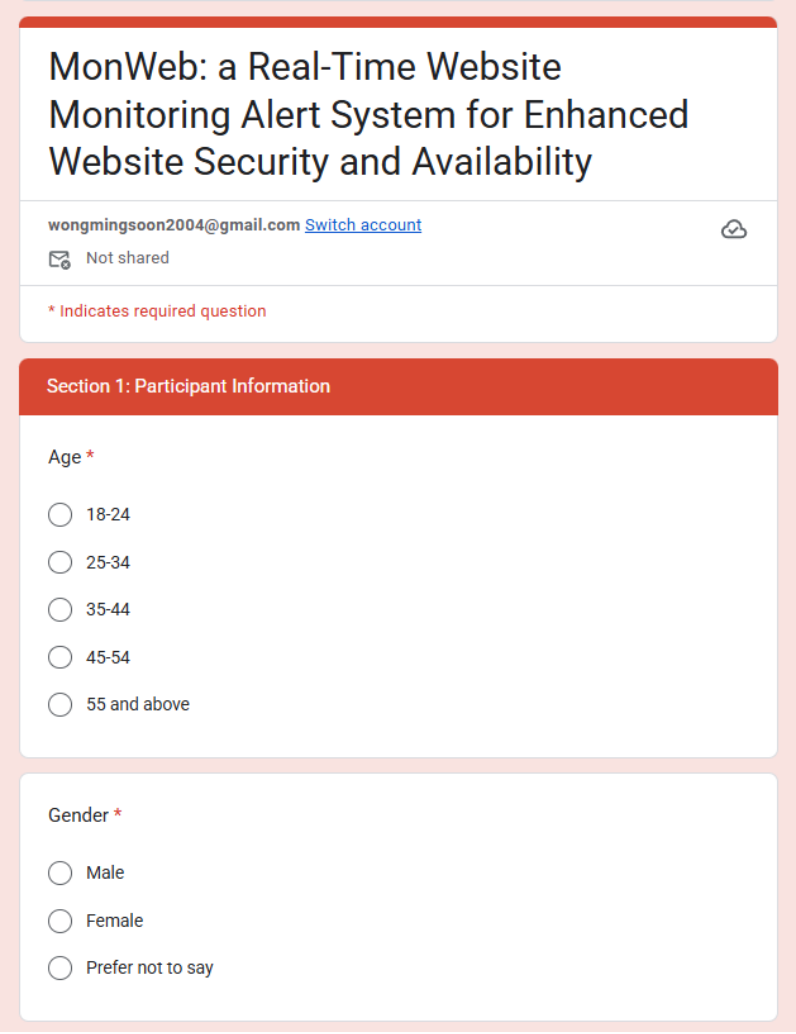


Figure 35: Section 1 of Survey's First Few Question

Next, after the participant has done page 1, which is the introduction page, the survey form will show the multiple-choice questions. The figure shows section 1 of the survey, which

is used to focus on collecting the participants' information. The aim of this section 1 is to collect the basic participant information, such as the age of the participant, gender, higher educational background, current professional role of the participant, and technical expertise. Recording and collecting the information can help to better understand the profile and professional level related to the MonWeb project from the respondents.

The use of the first few questions is to get the basic demographic questions, such as the age range (18-24, 25-34, 35-44, 45-54, 55 and above) and gender options (Male, Female, prefer not to say). These two questions can also help this survey complete the basic screening of the target population.

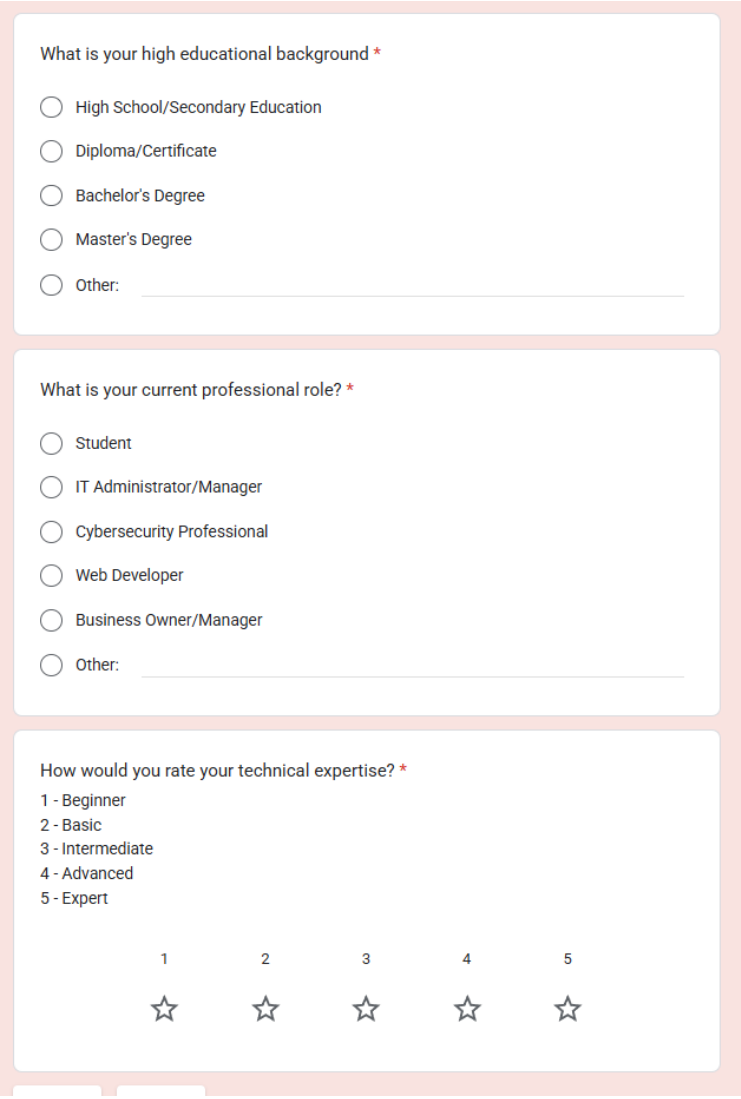


Figure 36: Section 1 of Survey's Last Few Questions

Next, it will need to get into the participant's educational background, including the High School, Diploma, Bachelor's Degree, and master's Degree. Additionally, further understanding the participants' current professional roles, including Students, IT Administrators or Managers, Cybersecurity Professionals, Web Developers, and Business Owners or Managers, is crucial. These questions can help analyse the reasons behind survey results from different educational backgrounds and professional groups, thereby generating a more authentic survey result and avoiding research bias.

Section 2: Requirement of Website Monitoring

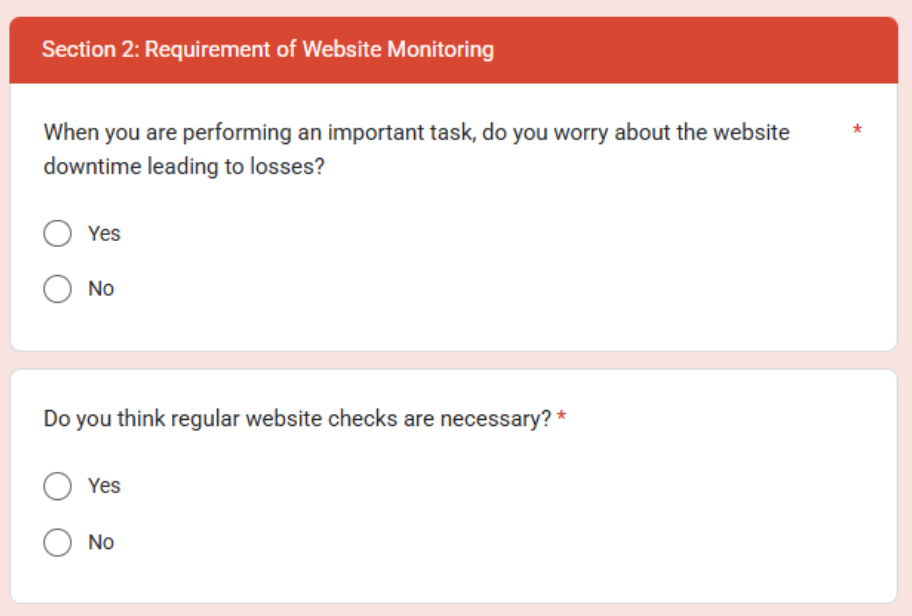


Figure 37: Section 2 of Survey's First Few Questions

The figure above shows the second section of the survey, which is "Section 2: Requirement of Website Monitoring". The goal of this section is to address the important concerns regarding website monitoring from participants.

The first few questions are used to understand the pattern of participants when facing the potential losses due to website downtime and views on the necessity of regular website checks. Participants will only need to select Yes or No. These questions can help to provide the opinion of participants facing any website's issue situation.

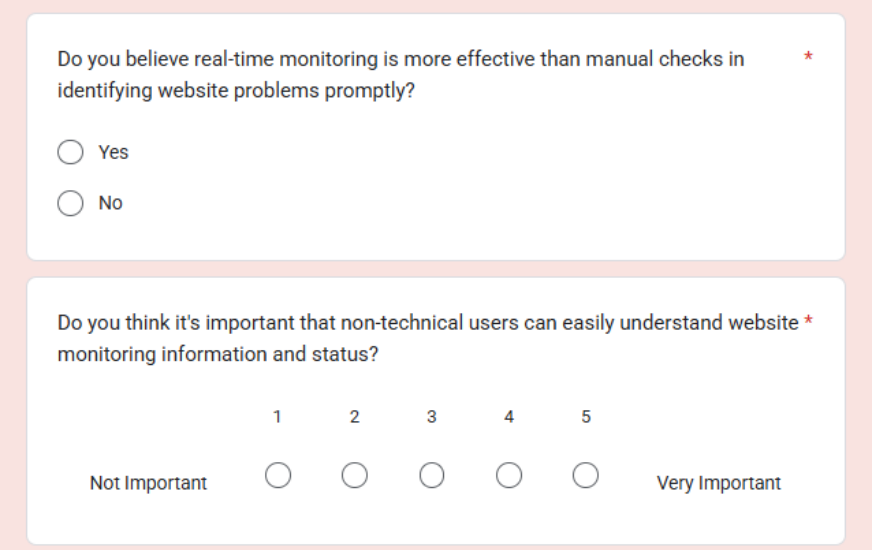


Figure 38: Section 2 of Survey's Middle Questions

Additionally, it investigates the effectiveness of website monitoring with manual checking, and it is important for the non-technical team to understand the basic knowledge of website monitoring. These questions are used to let participants know the importance of website monitoring on a normal website.

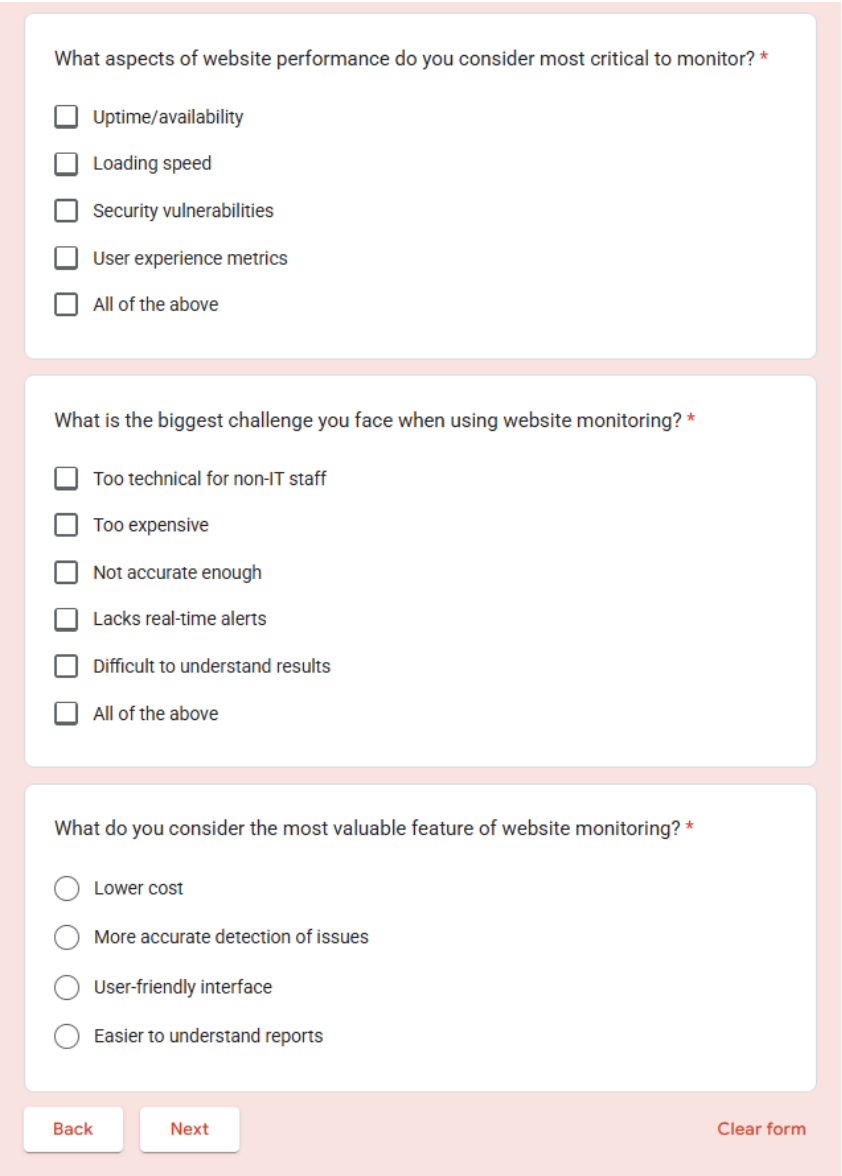


Figure 39: Section 2 of Survey's Last Few Questions

After that, the last few questions will get responses about the aspect of website performance and valuable features that participants want to have inside the website monitoring. Get to know the challenge or issues that the participant has been facing before. These questions can ensure the performance, function, and suggestions that can improve the website monitoring requirement.

Section 3: AI features in website monitoring

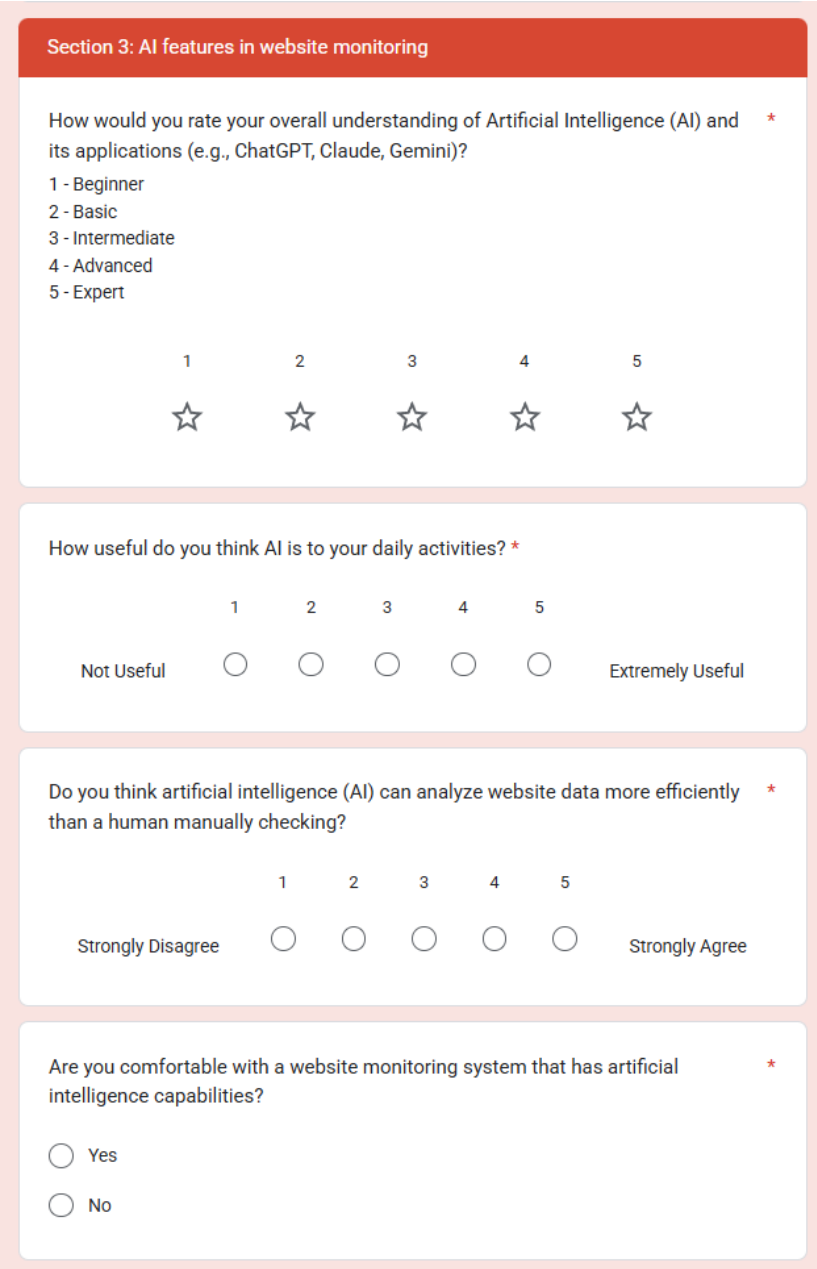


Figure 40: Section 3 of Survey's First Few Questions

The figure shows the third section of the survey, which is "Section 3: AI features in website monitoring". The goal of this section is to collect data about the effectiveness of different monitoring approaches and options for website performance aspects.

The first few questions focus on the understanding, usefulness, efficiency, and comfort of Artificial Intelligence that integrates into the website monitoring. These questions can assess

whether participants can accept AI as a tool with practical value in their daily lives. Additionally, they can investigate whether participants agree that AI is a tool that can improve efficiency compared to manually completing daily tasks.

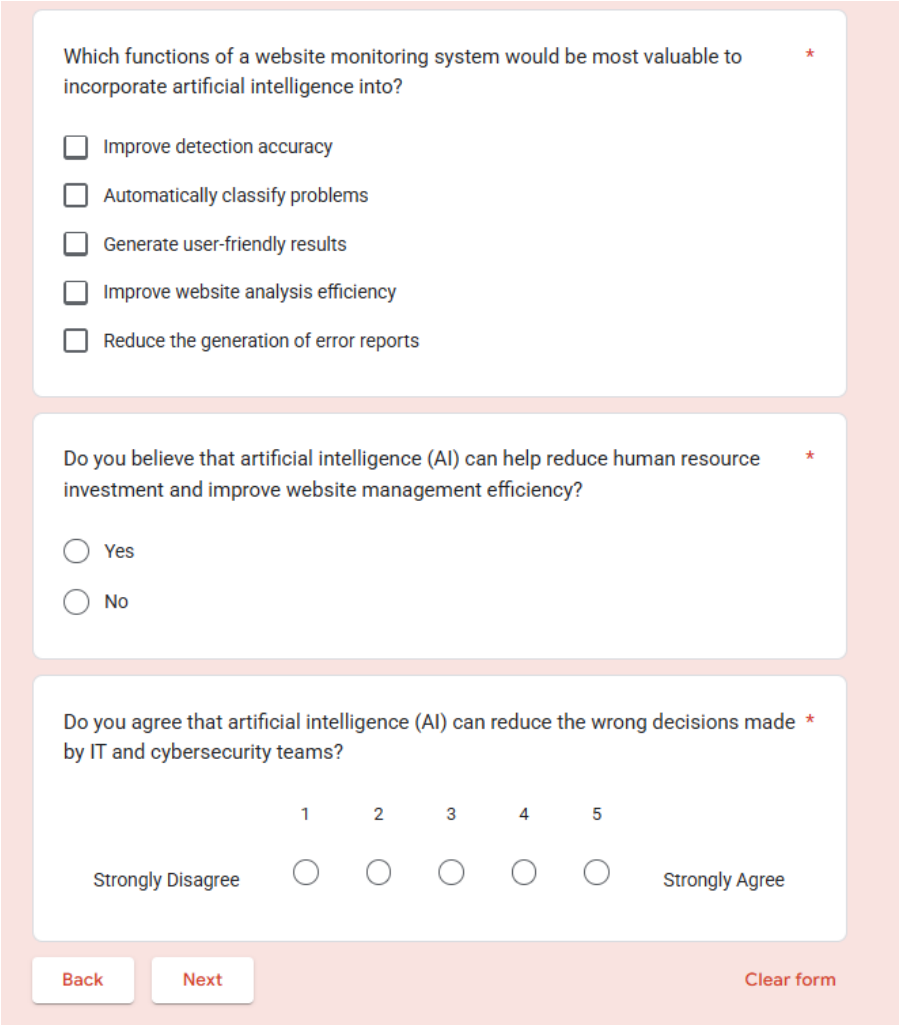


Figure 41: Section 3 of Survey's Last Few Questions

In addition, the last few questions are about the expectation from participants that functions can increase the usefulness of website monitoring and enhance the confidence in the functionality of the artificial intelligence.

Section 4: Validation of the user requirements of visualization and report generation

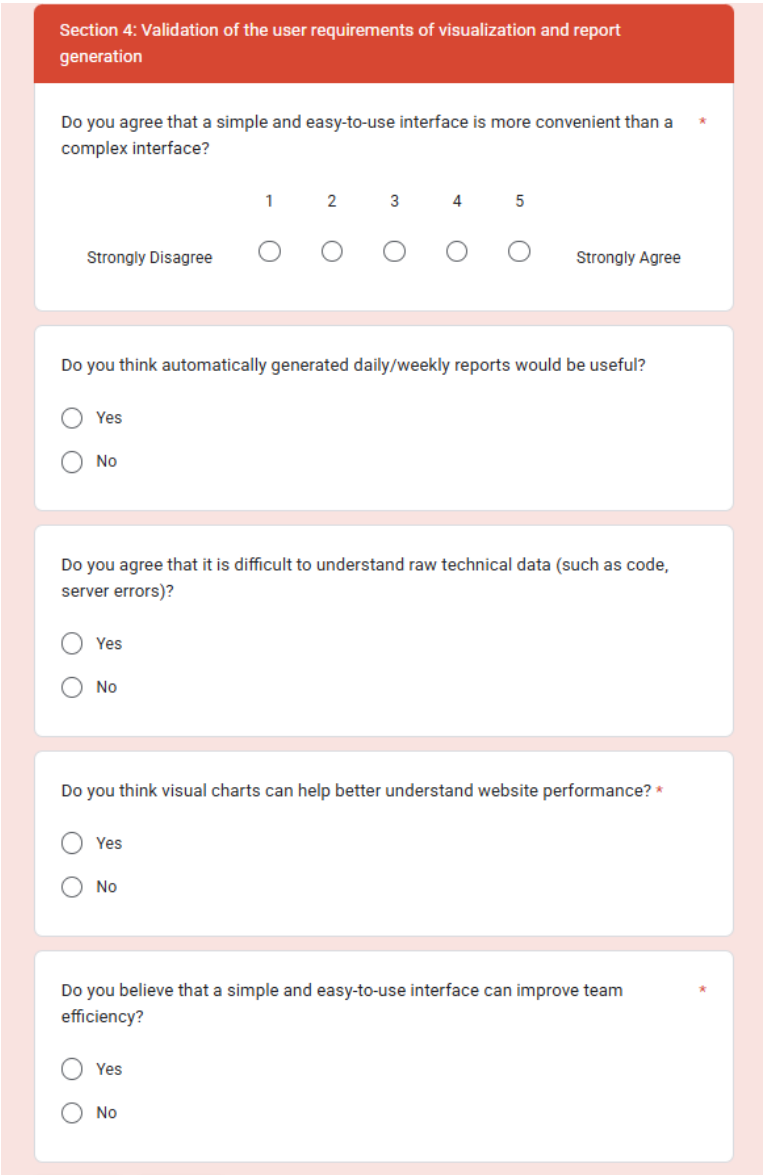


Figure 42: Section 4 of Survey's First Few Questions

The figure shows the third section of the survey, which is "Section 4: Validation of the user requirements of visualisation and report generation". The goal of this section 4 is to collect the requirements of interface usability, reporting functionality, and data visualisation effectiveness from participants' feedback.

The first of the few questions will be used to show that the feedback about a simple and easy-to-use user interface can increase the efficiency, usefulness of automation in generating reports, and the difficulty of raw technical data. These questions can highlight the importance

of the real value of using a user-friendly interface. The last few questions investigate how the performance of the visual chart and user-friendly interface can improve the team's efficiency.

Section 5: Closing

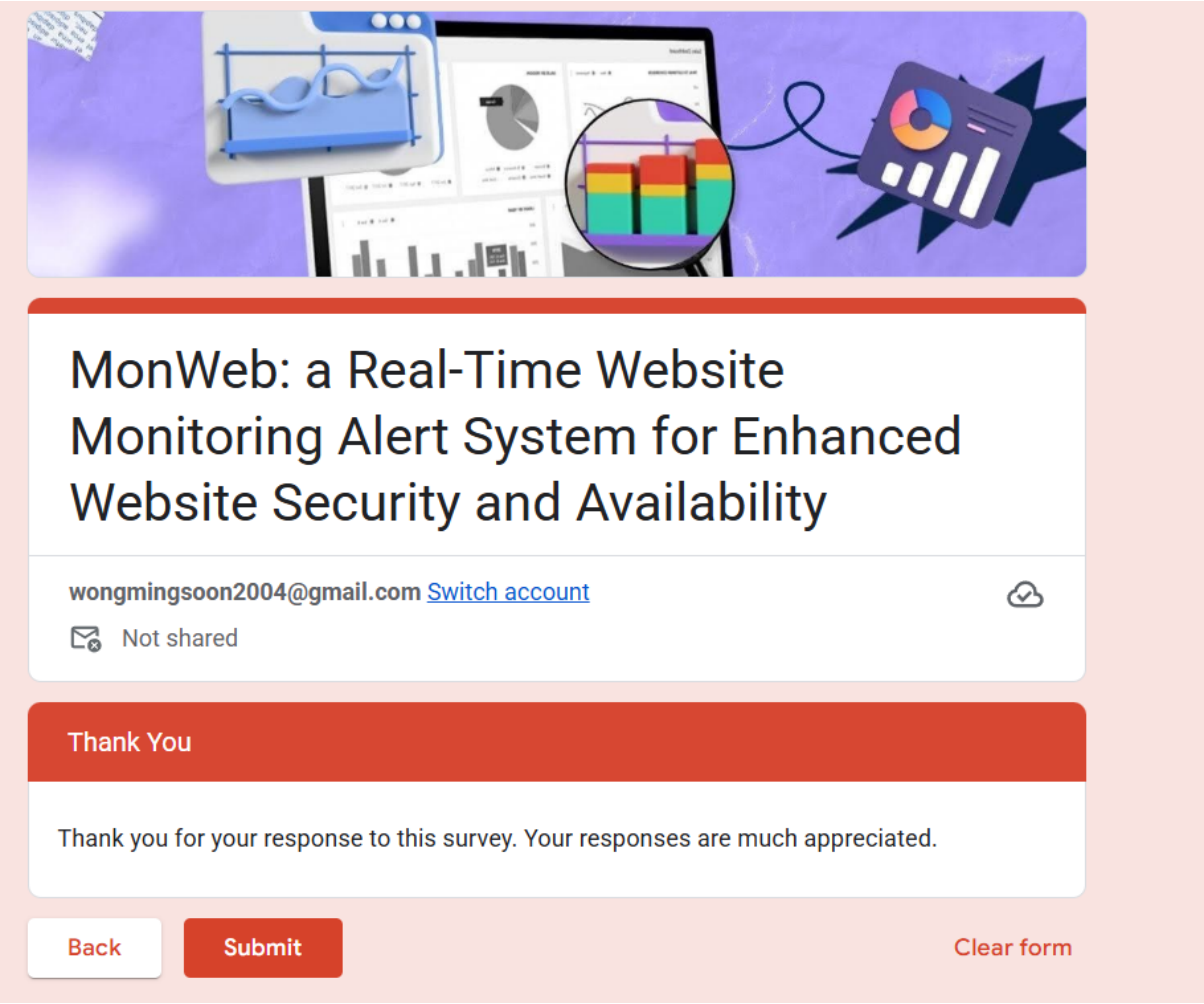


Figure 43: Section 5 of Survey's Closing

This section is the last page of the survey, which is to show the thanks page for all the participants. The aim of this section is to appreciate participants for completing and responding to the survey.

3.3 Analysis

Section 1: Participant Information

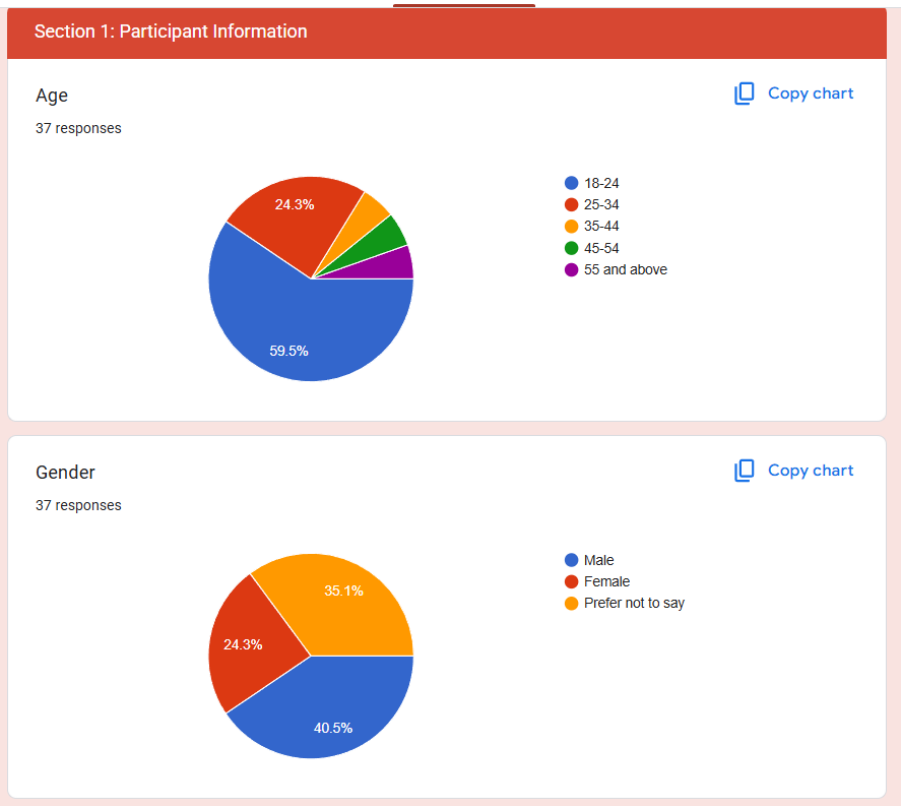


Figure 44: Survey Result of Age and Gender

The figure above shows the results of participants' age and gender. Most of the age is around 18-24, and the second highest range of the age is 25-34. Many participants were male, and the second largest group of participants chose to prefer not to say.

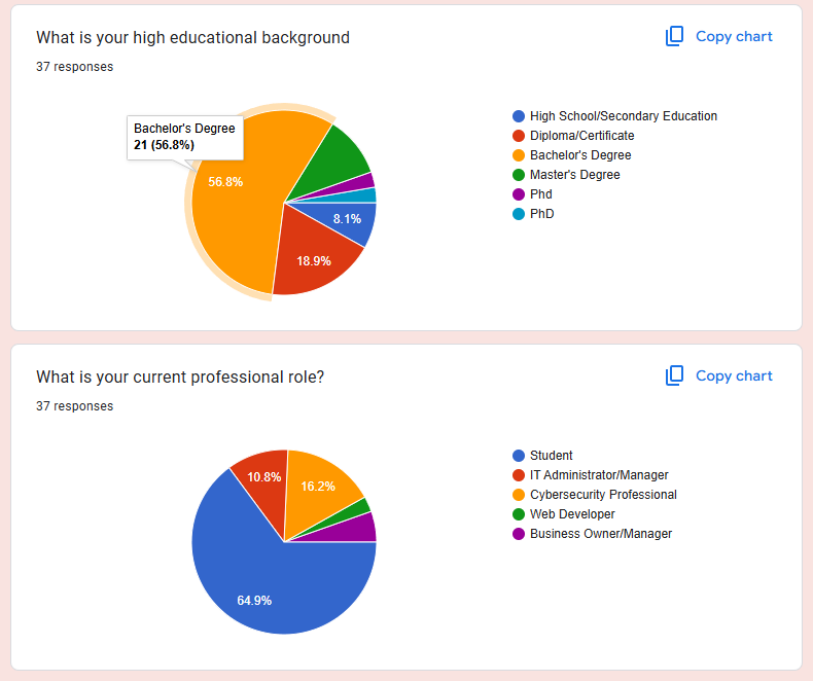


Figure 45: Survey Result of Educational Background and Professional Role

The next question about the participant information is the higher educational background and the current participant's professional role. Most of the participants have a higher educational background, with a bachelor's degree, and the second highest is a Diploma or Certificate.

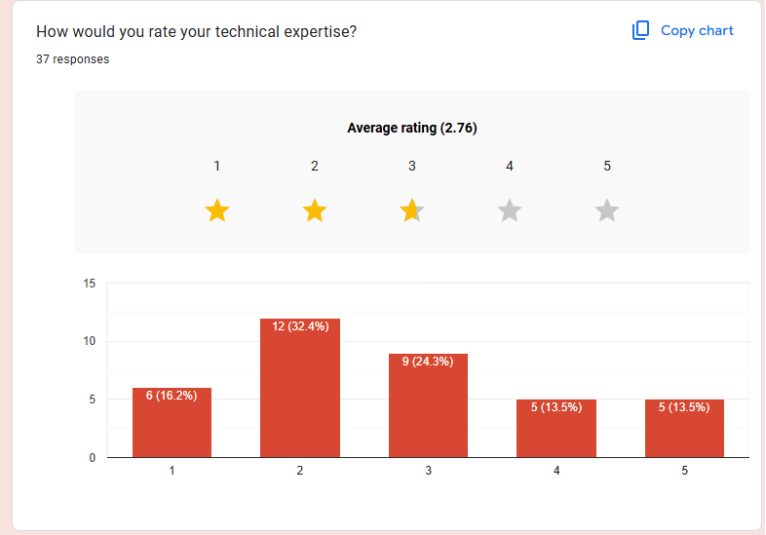


Figure 46: Survey Result of Technical Expertise

This question shows the technical expertise of participants, where the options are beginner, basic, intermediate, advanced, and expert (1-5). Most of the technical expertise of participant is at a basic level, which means they have some foundation knowledge and skill in their technical field, but do not have professional skills in technical, such as principal technical and formulation technical.

Section 2: Requirement of Website Monitoring

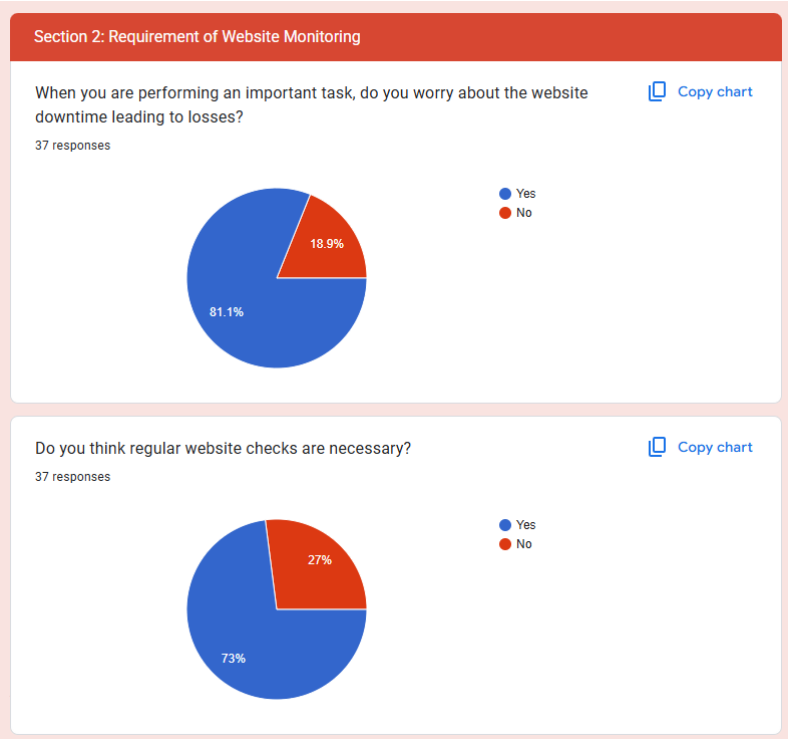


Figure 47: Survey Result of Website Checking Understanding

Next, these questions are the first few questions from section 2. 81.1% of 37 responses indicate that they are worried about the website downtime leading to losses when they are using the website to perform important tasks, such as the bank website, top-up website, login page, and item viewing page. Around 73% of responses agree that website checks are necessary to ensure website security.

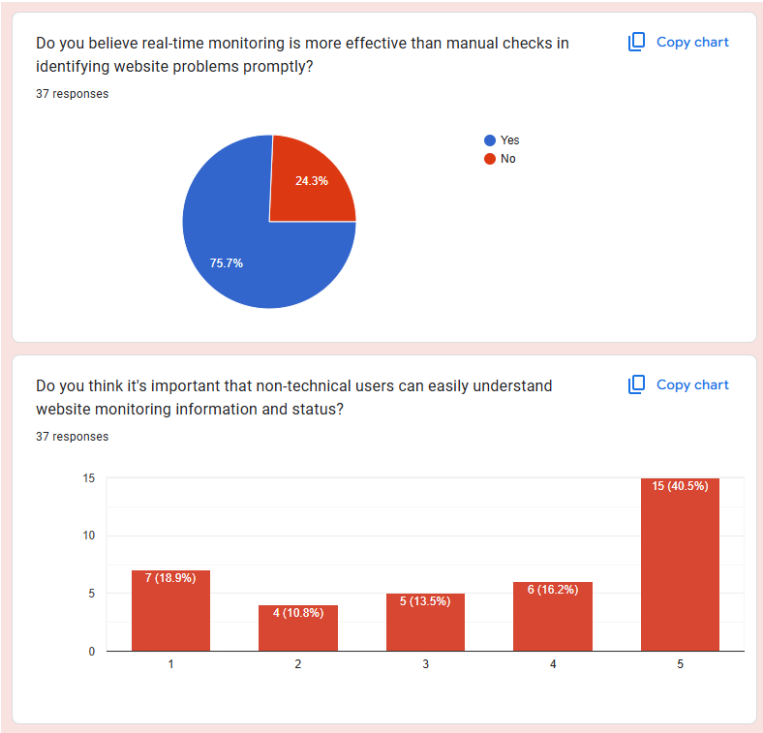


Figure 48: Survey Result of Section 2's Middle Questions

Around 75.7% of the 37 responses agree that real-time monitoring is more effective than manual checking in identifying website problems. The next question shows the importance of non-technical users understanding the website monitoring information and status, where the options are from not important to very important (1-5). More than half of the participants gave responses that non-technical users should understand the website monitoring information and status.



Figure 49: Survey Result of Expected Website Monitoring Performance

The first question of the figure above shows the most critical parts of the website that need to be identified, which are uptime or availability, loading speed, security vulnerabilities, and user experience metrics. According to the participants' feedback, the availability of the website is an important part that needs to be focused on, and around 40.5% think that user experience metrics also must be covered in monitoring. The next question's result is the pricing of the website monitoring tools, and the high requirement for non-IT staff is the biggest challenge for participants. Well, the lowest challenge participants face when using website monitoring is that it is not accurate enough.

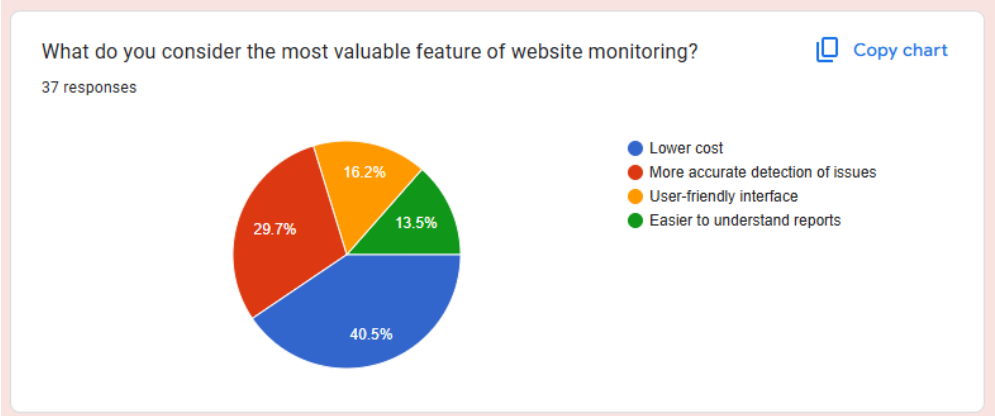


Figure 50: Survey Result of The Last Question of Section 2

Lower cost is the most preferred feature of website monitoring from the above figure. The most valuable feature of website monitoring is that it makes reports easier to understand. From the result of the figure above, it means lower cost and more accurate detection of issues are the participants' focus on the valuable feature.

Section 3: AI features in website monitoring

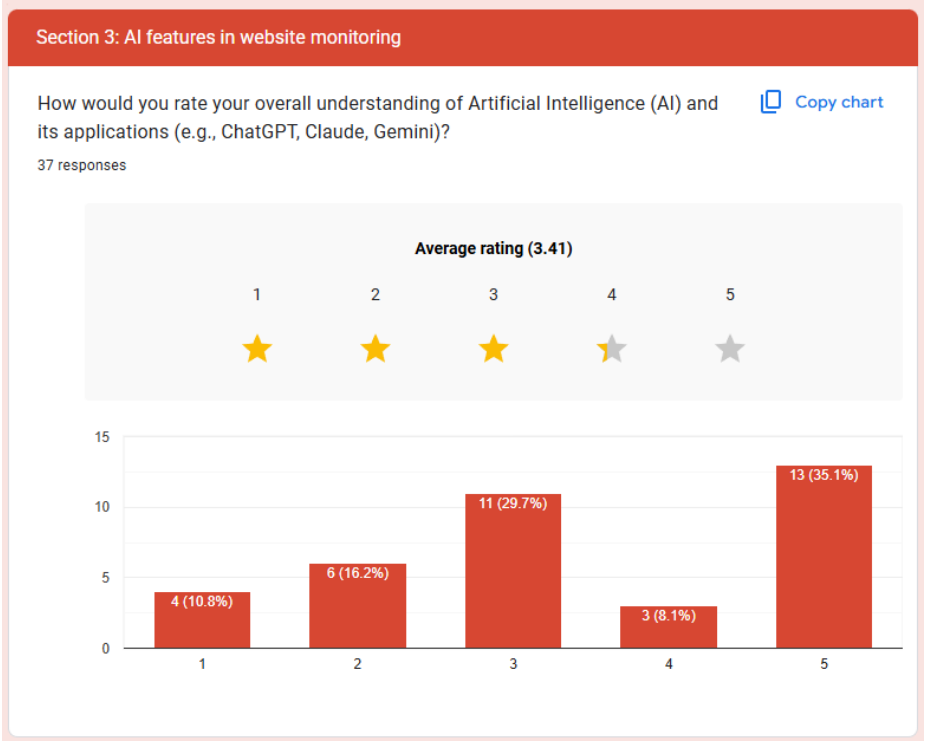


Figure 51: The First Question of Section 3

The figure above shows the interesting patterns and understanding of artificial intelligence among participants. The average rating is 3.41 out of 5, which means the overall knowledge level is intermediate. According to the survey result, the highest percentage is 35.1% which means that 13 participants think that they are at an expert level. Then, the next highest percentage is 29.7% which means that 11 participants think that they are on an intermediate level. This survey result shows that their knowledge level of artificial intelligence expertise is diverse. Some of the participants feel that they are masters of the subject or are still improving their knowledge level, but fewer of the participants feel that their knowledge level range is lower.



Figure 52: Survey Result of Artificial Intelligence's Understanding of Section 3

The figure above is more focused on the understanding of artificial intelligence from participants, such as practicality and the perception of website monitoring efficiency. The highest percentage of the first and two questions is the same, which is 45.9% and 17 participants are respondents for it. The summary of the first question's result is that 12 participants feel that artificial intelligence is not useful to their daily activities, while over half of the participants agree that artificial intelligence is useful to their daily activities, showing significant cognitive

differences within the group. Next, the summary of the second question's result is that nearly half of the group highly agree with the actual effectiveness of artificial intelligence.

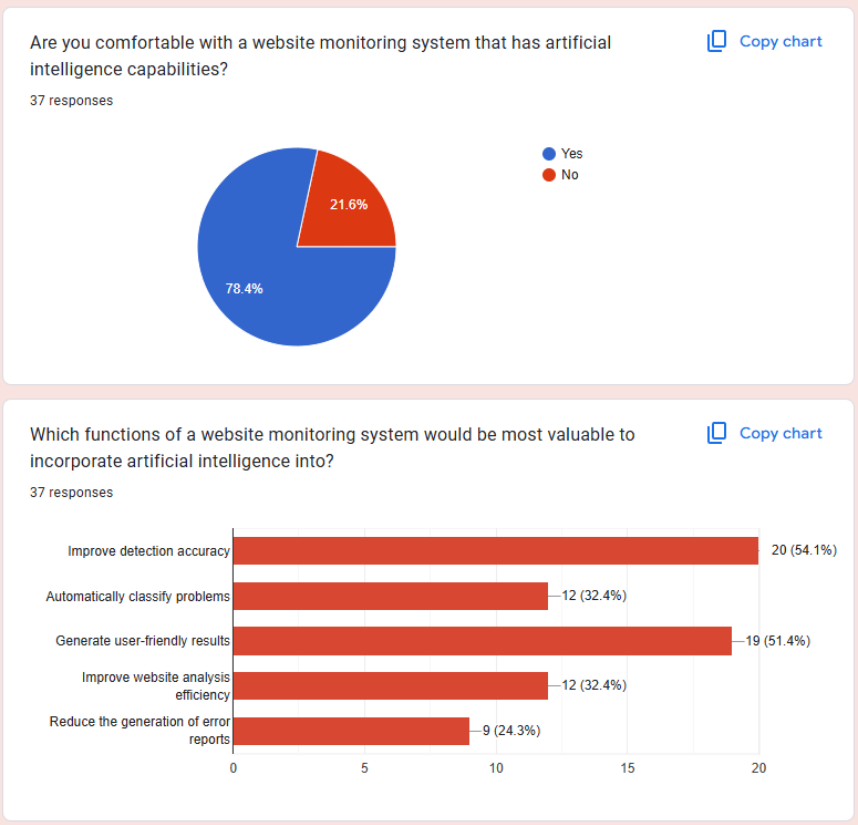


Figure 53: Survey Result of Expected Website Monitoring's Valuable Functions

According to the chart's figure from the question's result, it used to collect the data about the user attitude and pattern from the artificial intelligence features of website monitoring systems. The summary of the first question's result is that 78.4% feel comfortable with website monitoring integrating artificial intelligence capabilities, while 21.6% feel uncomfortable, which means most of the participants have a positive attitude toward artificial intelligence technology in website monitoring. Next, the bar chart above it was used to explore the features of a website monitoring system that are most valuable to incorporate artificial intelligence. The options of improving detection accuracy and generating user-friendly results are the highest supported by participants. The summary of the second question's result is that users value artificial intelligence most in improving system accuracy and increasing the user experience.

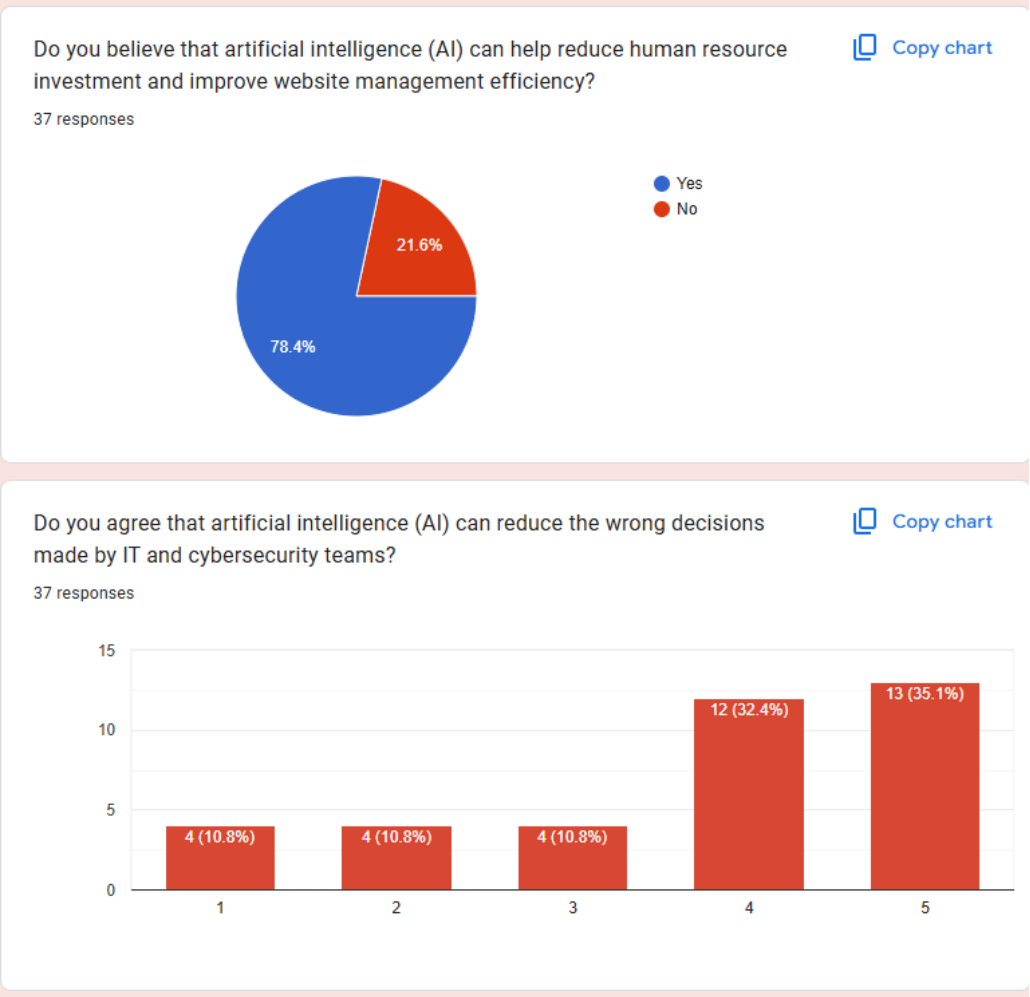


Figure 54: Survey Result of Artificial Intelligence's Benefit

The figure above shows the survey chart's result, which is used to investigate whether artificial intelligence can help to reduce human resource investment and improve the website management efficiency from participants. The summary of this question's result is that most of the participants believe that artificial intelligence can bring benefits, such as increasing efficiency and reducing resource investment. In addition, the second question shows that many participants agree that artificial intelligence can help to improve management efficiency and reduce the wrong decision situation.

Section 4: Validation of the user requirements of visualization and report generation

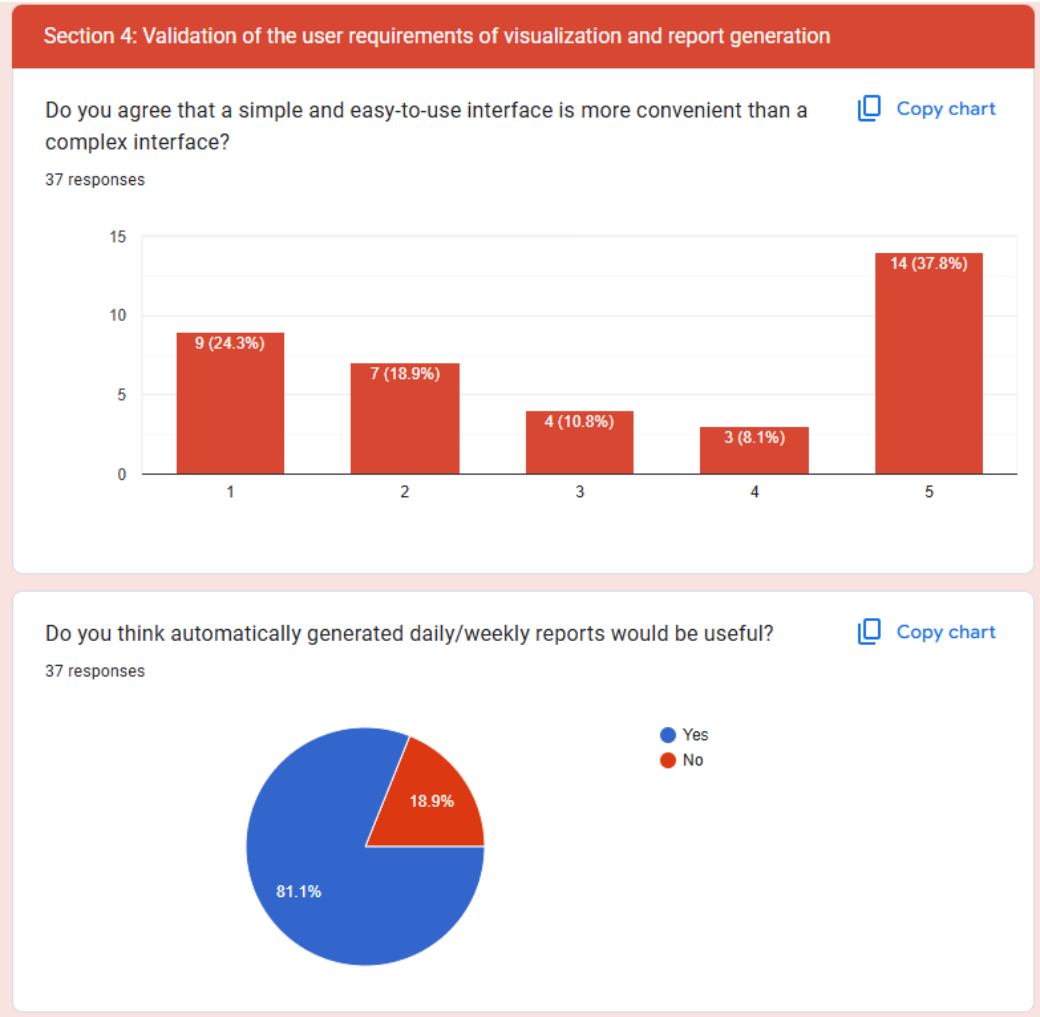


Figure 55: Survey Result of Simple Interface's opinion

From the figure above, the questions begin in Section 4: Validation of the user requirements of visualisation and report generation. The first question is used to investigate whether participants agree that a simple and easy-to-use interface is more convenient than a complex interface. The results show that the agree option is the most selected, around 37.8% 14 respondents. The second question is to understand whether users find automatically generated daily or weekly reports useful. The summary of this question is that 81.1% of 30 respondents agree that this feature is useful.

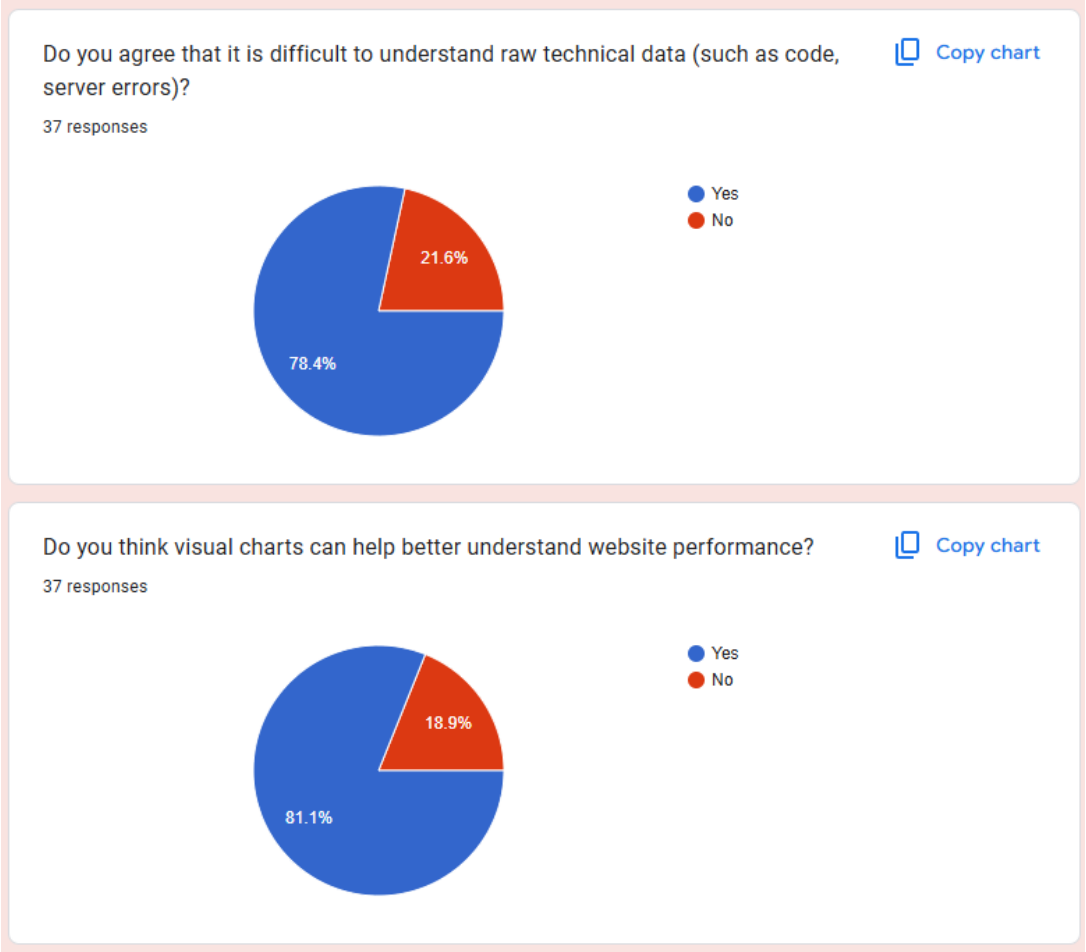


Figure 56: Survey Result of Participant's Opinion on non-technical Person

The figure above shows two pie charts about the understanding and visualisation of technical data from participants. The summary of the first question is that most of the participants agree with the question, 78.4%. The summary of the second question is that most of the participants agree that a visual chart can help to better understand the website performance, with around 81.1% of participants. It has been shown that the raw technical data is difficult to understand for the majority of participants, and the importance of data visualisation in transferring technical information.

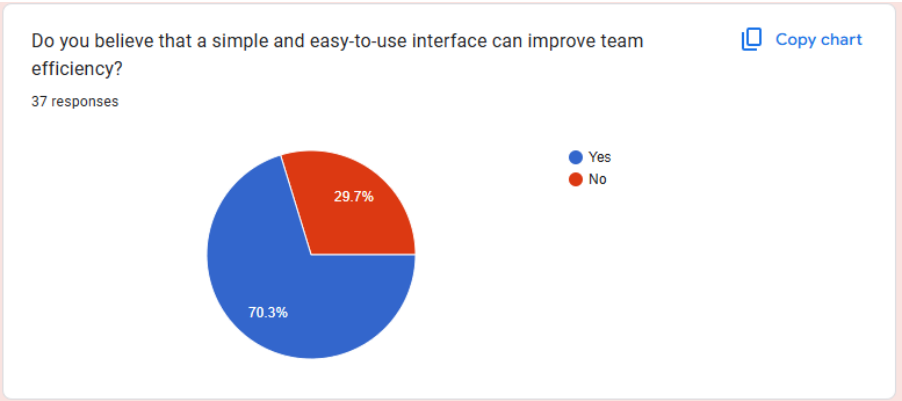


Figure 57: Survey Result of the Last Question

The figure above show majority of the participant select to believe the simple and easy-to-use can enhance their efficiency.

3.4 Summary of Analysis

This survey collected 37 responses from participants, with the age range of most participants being 18-24, primarily from students with a computing background, and most of the participants are around bachelor's degree and diploma students. In addition, some of the participants have a technical background, but still have limitations in website monitoring systems and artificial intelligence. Although most of the participants came from different educational backgrounds and current professional roles, they still have different knowledge levels of artificial intelligence and natural language processing in website monitoring systems. It will show that the survey result shows a knowledge gap between participants and advanced monitoring technologies.

Not only that, but participants' also focus on websites is gradually increasing; therefore, the main focus of this survey is on users' demand for real-time website detection functions, the necessity of AI-integrated detection systems, and the development of user-friendly system interfaces centred on non-technical users. Although many participants lack specialised technical knowledge and skills, they still recognise the importance and benefits of having a complete detection mechanism for important websites, adding AI to enhance accuracy, automating the generation of easy-to-understand detection reports, and building flexible and accessible dashboards.

CHAPTER 4: CONCLUSION

4.1 Achievement

The first part of the project successfully established a full foundation for the MonWeb system, a real-time website monitoring and alert system, which was achieved through the literature review and data collection of a survey to make a deeper analysis of the weaknesses of the existing website monitoring. It can also have a clear focus on promoting SDG 9: Industry, Innovation and Infrastructure. The key issues validation includes the limitation of website detection, the lack of a friendly interface for non-technical users, and the limitations of functions for fixing complex websites. In the research part, there will be a deeper domain research on Artificial Intelligence (AI), Natural Language Processing (NLP), and website monitoring techniques to ensure the relevance to industry advanced systems and identify their strength and weaknesses.

4.2 Adequate Research

The investigation and research part can provide a clear direction for achieving the objective of the MonWeb system, a real-time website monitoring and alert system with SDG 9: Industry, Innovation, and Infrastructure. According to the usable resources and research, it can support verifying the design and development strategy, thereby providing a strong theoretical foundation. The technical research can provide confirmation of the tools and techniques that can balance the feasibility with innovation. The data collection, which is the survey gathering data, can ensure the relevant valuable data on the preferences and behaviours of the users.

4.3 Gaps and Future

Although the research and investigation of the project have provided a comprehensive foundation, some gaps still need to be further explored and improved. One of the improvements is the technical implementation, which is the Artificial Intelligence model optimisation. The current investigation of the technical research is based on the pre-existing datasets, which will have limitations in technical aspects for design and development. By developing a collection and analysing the real-time user data, we can fix the technical limitations.

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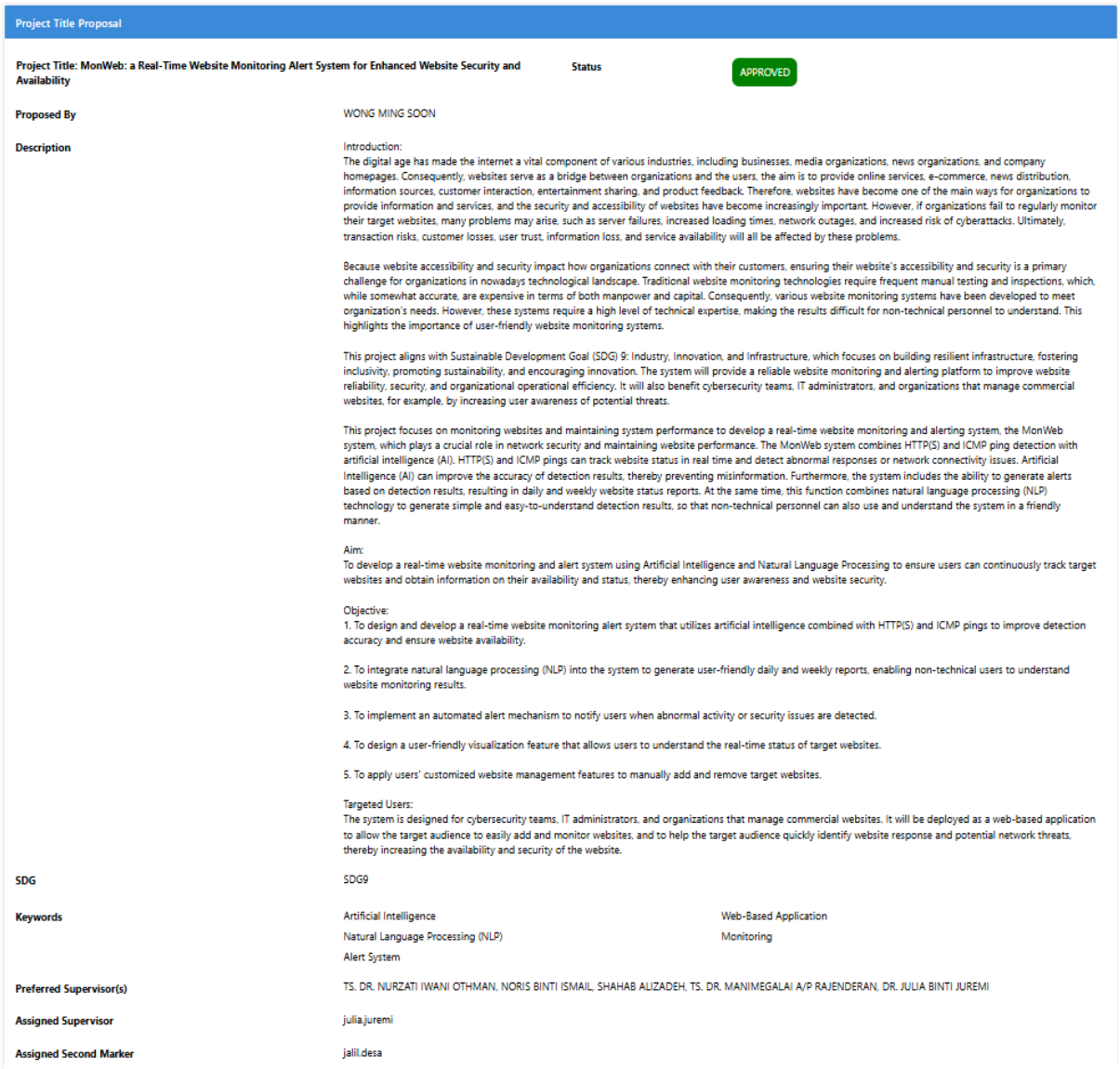
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APPENDICES

Appendix A: PPF – Title Registration Proposal



Project Title Proposal

Project Title: MonWeb: a Real-Time Website Monitoring Alert System for Enhanced Website Security and Availability	**Status** APPROVED
Proposed By	WONG MING SOON
Description	Introduction: The digital age has made the internet a vital component of various industries, including businesses, media organizations, news organizations, and company homepages. Consequently, websites serve as a bridge between organizations and the users, the aim is to provide online services, e-commerce, news distribution, information sources, customer interaction, entertainment sharing, and product feedback. Therefore, websites have become one of the main ways for organizations to provide information and services, and the security and accessibility of websites have become increasingly important. However, if organizations fail to regularly monitor their target websites, many problems may arise, such as server failures, increased loading times, network outages, and increased risk of cyberattacks. Ultimately, transaction risks, customer losses, user trust, information loss, and service availability will all be affected by these problems. Because website accessibility and security impact how organizations connect with their customers, ensuring their website's accessibility and security is a primary challenge for organizations in nowadays technological landscape. Traditional website monitoring technologies require frequent manual testing and inspections, which, while somewhat accurate, are expensive in terms of both manpower and capital. Consequently, various website monitoring systems have been developed to meet organization's needs. However, these systems require a high level of technical expertise, making the results difficult for non-technical personnel to understand. This highlights the importance of user-friendly website monitoring systems. This project aligns with Sustainable Development Goal (SDG) 9: Industry, Innovation, and Infrastructure, which focuses on building resilient infrastructure, fostering inclusivity, promoting sustainability, and encouraging innovation. The system will provide a reliable website monitoring and alerting platform to improve website reliability, security, and organizational operational efficiency. It will also benefit cybersecurity teams, IT administrators, and organizations that manage commercial websites, for example, by increasing user awareness of potential threats. This project focuses on monitoring websites and maintaining system performance to develop a real-time website monitoring and alerting system, the MonWeb system, which plays a crucial role in network security and maintaining website performance. The MonWeb system combines HTTP(S) and ICMP ping detection with artificial intelligence (AI). HTTP(S) and ICMP pings can track website status in real time and detect abnormal responses or network connectivity issues. Artificial Intelligence (AI) can improve the accuracy of detection results, thereby preventing misinformation. Furthermore, the system includes the ability to generate alerts based on detection results, resulting in daily and weekly website status reports. At the same time, this function combines natural language processing (NLP) technology to generate simple and easy-to-understand detection results, so that non-technical personnel can also use and understand the system in a friendly manner. Aim: To develop a real-time website monitoring and alert system using Artificial Intelligence and Natural Language Processing to ensure users can continuously track target websites and obtain information on their availability and status, thereby enhancing user awareness and website security. Objective: 1. To design and develop a real-time website monitoring alert system that utilizes artificial intelligence combined with HTTP(S) and ICMP pings to improve detection accuracy and ensure website availability. 2. To integrate natural language processing (NLP) into the system to generate user-friendly daily and weekly reports, enabling non-technical users to understand website monitoring results. 3. To implement an automated alert mechanism to notify users when abnormal activity or security issues are detected. 4. To design a user-friendly visualization feature that allows users to understand the real-time status of target websites. 5. To apply users' customized website management features to manually add and remove target websites. Targeted Users: The system is designed for cybersecurity teams, IT administrators, and organizations that manage commercial websites. It will be deployed as a web-based application to allow the target audience to easily add and monitor websites, and to help the target audience quickly identify website response and potential network threats, thereby increasing the availability and security of the website.
SDG	SDG9
Keywords	Artificial Intelligence Natural Language Processing (NLP) Alert System Web-Based Application Monitoring
Preferred Supervisor(s)	TS. DR. NURZATI IWANI OTHMAN, NORIS BINTI ISMAIL, SHAHAB ALIZADEH, TS. DR. MANIMEGALAI A/P RAJENDERAN, DR. JULIA BINTI JUREMI
Assigned Supervisor	julia.juremi
Assigned Second Marker	jalil.desa

Figure 58: Title Registration Proposal

Appendix B: Ethics Forms

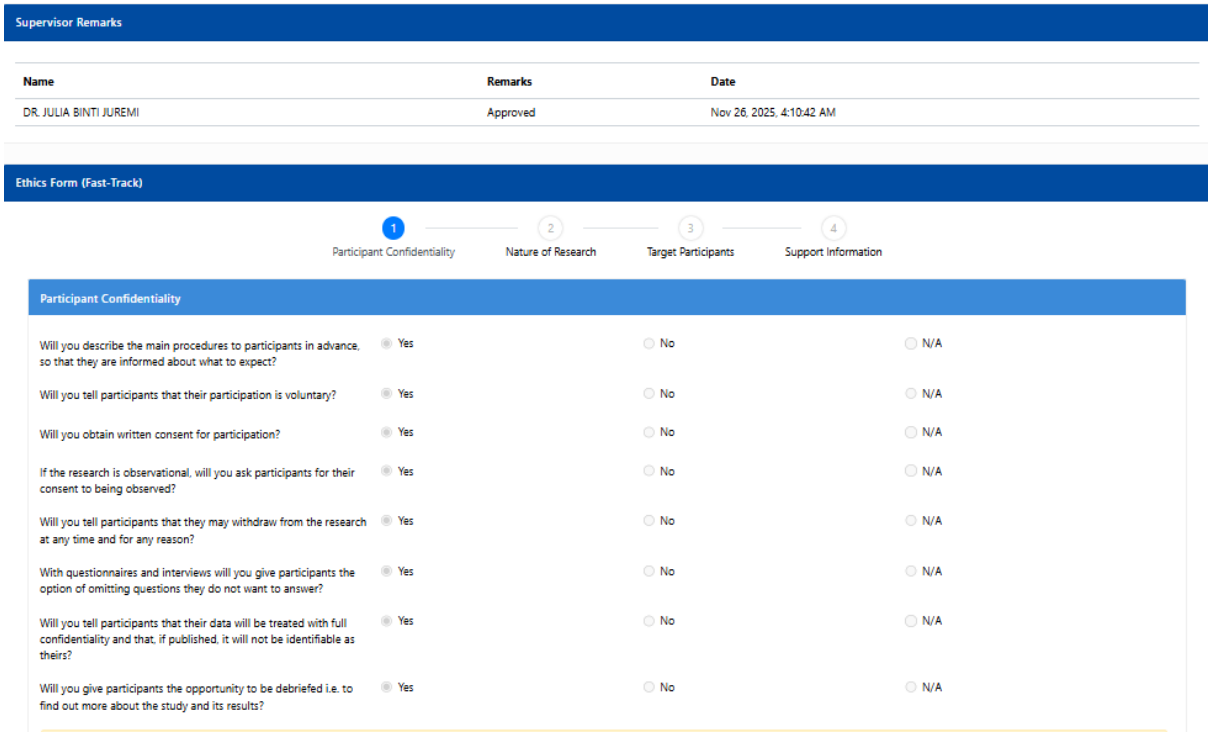


Figure 59: Ethics Forms

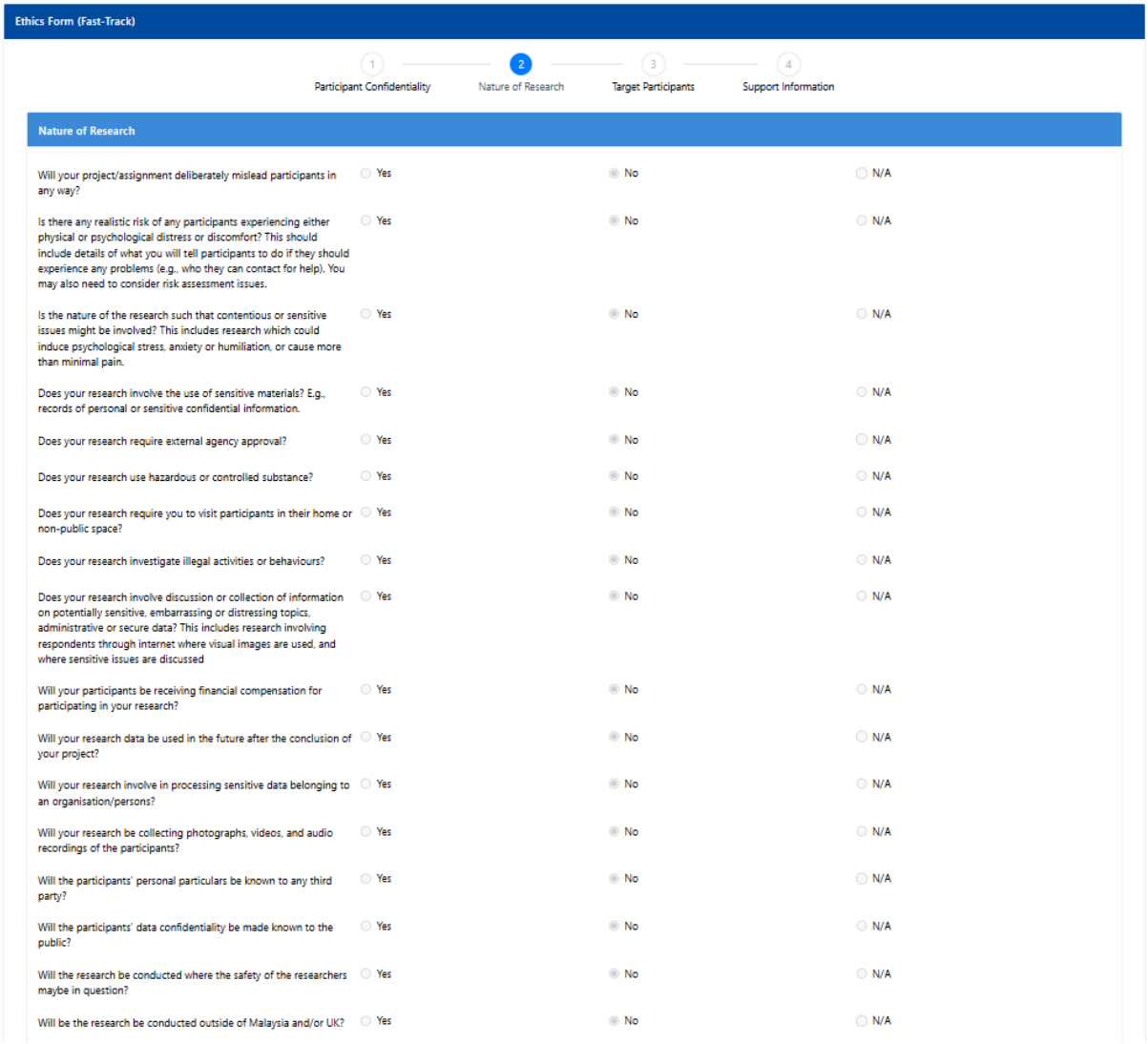


Figure 60: Nature of Research Ethics Forms

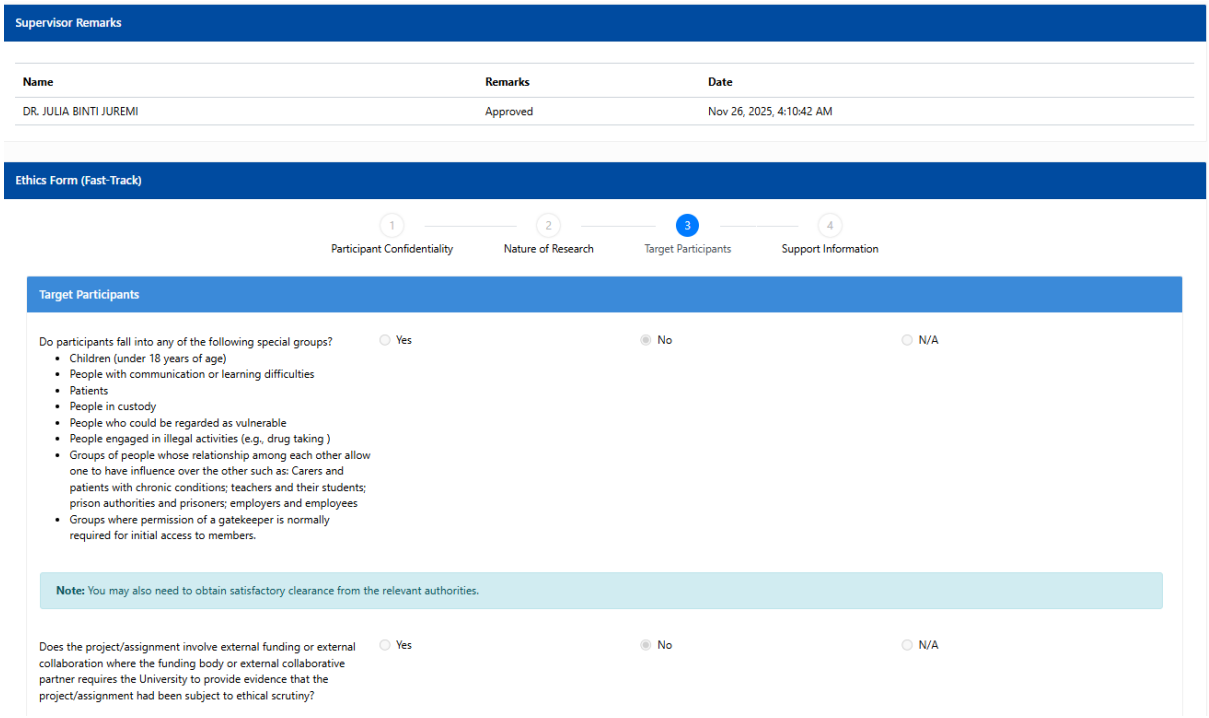


Figure 61: Target Participant of Ethics Forms

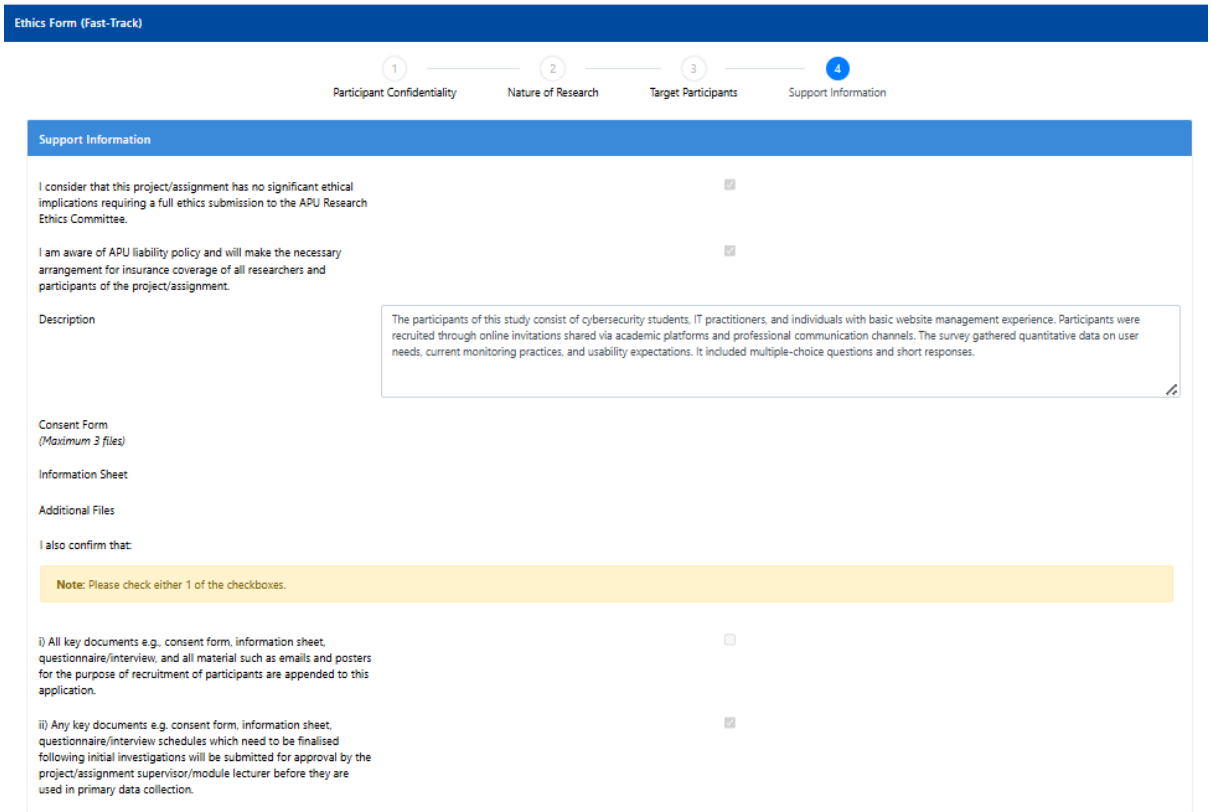


Figure 62: Support Information of Ethics Form

Appendix C: Log Sheets

Appendix C1: Log Sheets

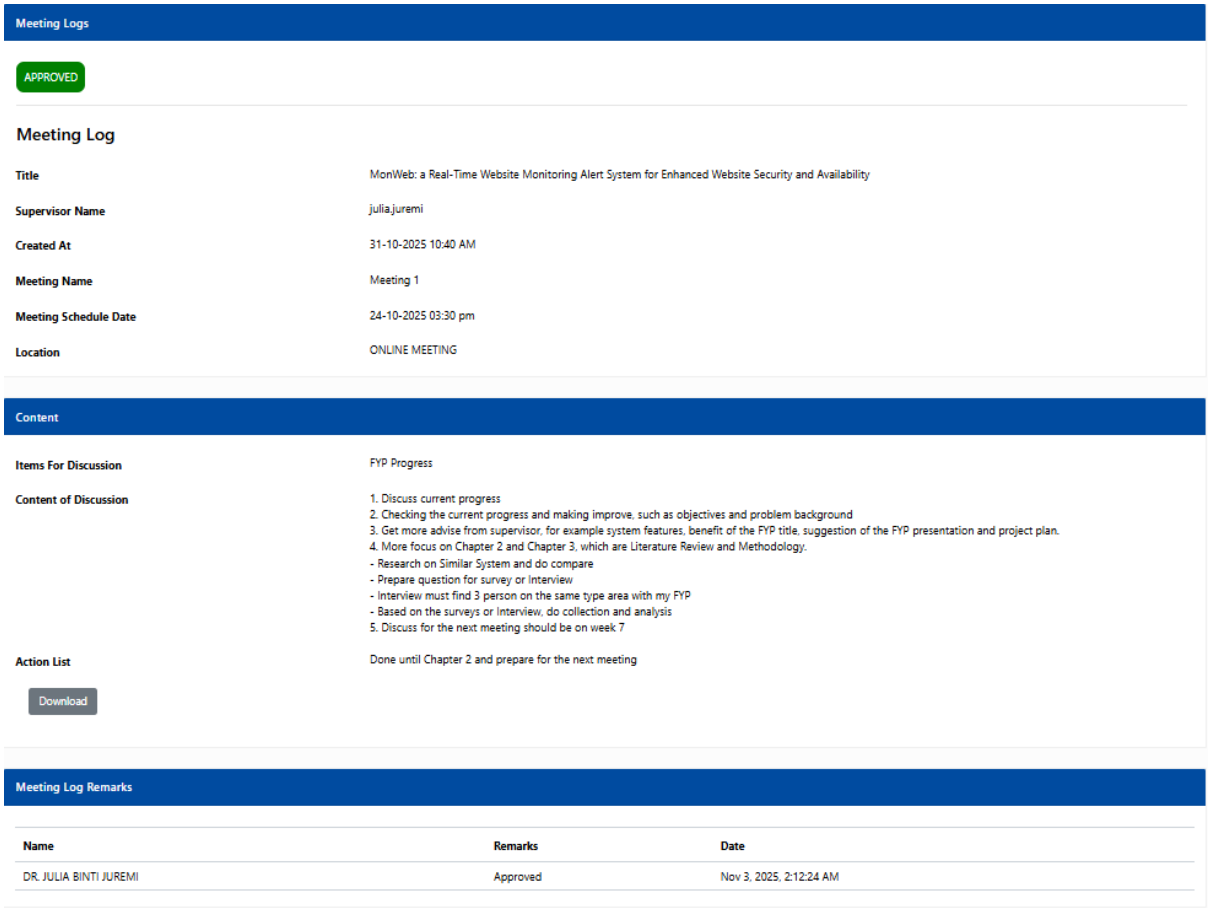


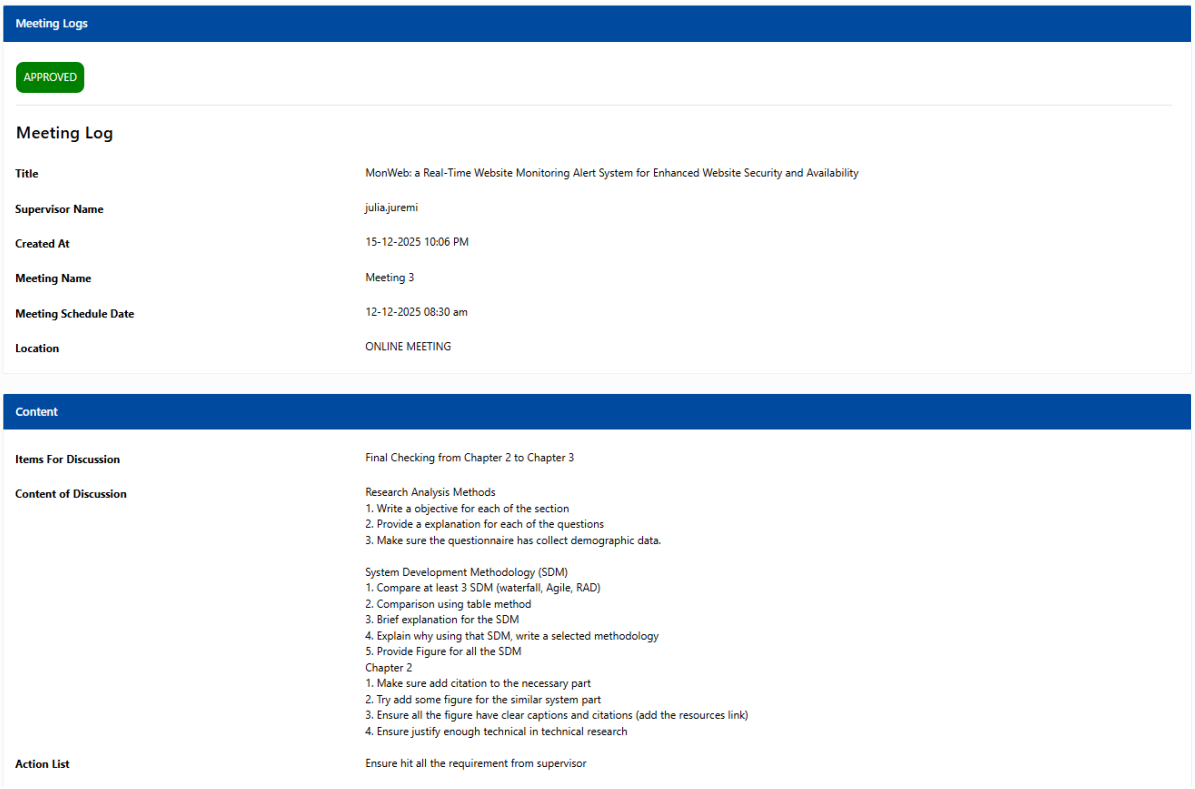
Figure 63: Meeting 1 Log Sheets

Appendix C2: Log Sheets



Figure 64: Meeting 2 Log Sheet

Appendix C3: Log Sheets



Meeting Logs

APPROVED

Meeting Log

Title	MonWeb: a Real-Time Website Monitoring Alert System for Enhanced Website Security and Availability
Supervisor Name	julia.juremi
Created At	15-12-2025 10:06 PM
Meeting Name	Meeting 3
Meeting Schedule Date	12-12-2025 08:30 am
Location	ONLINE MEETING

Content

Items For Discussion	Final Checking from Chapter 2 to Chapter 3
Content of Discussion	Research Analysis Methods 1. Write a objective for each of the section 2. Provide a explanation for each of the questions 3. Make sure the questionnaire has collect demographic data. System Development Methodology (SDM) 1. Compare at least 3 SDM (waterfall, Agile, RAD) 2. Comparison using table method 3. Brief explanation for the SDM 4. Explain why using that SDM, write a selected methodology 5. Provide Figure for all the SDM Chapter 2 1. Make sure add citation to the necessary part 2. Try add some figure for the similar system part 3. Ensure all the figure have clear captions and citations (add the resources link) 4. Ensure justify enough technical in technical research
Action List	Ensure hit all the requirement from supervisor

Figure 65: Meeting 3 Log Sheet

Appendix D: Gantt Chart

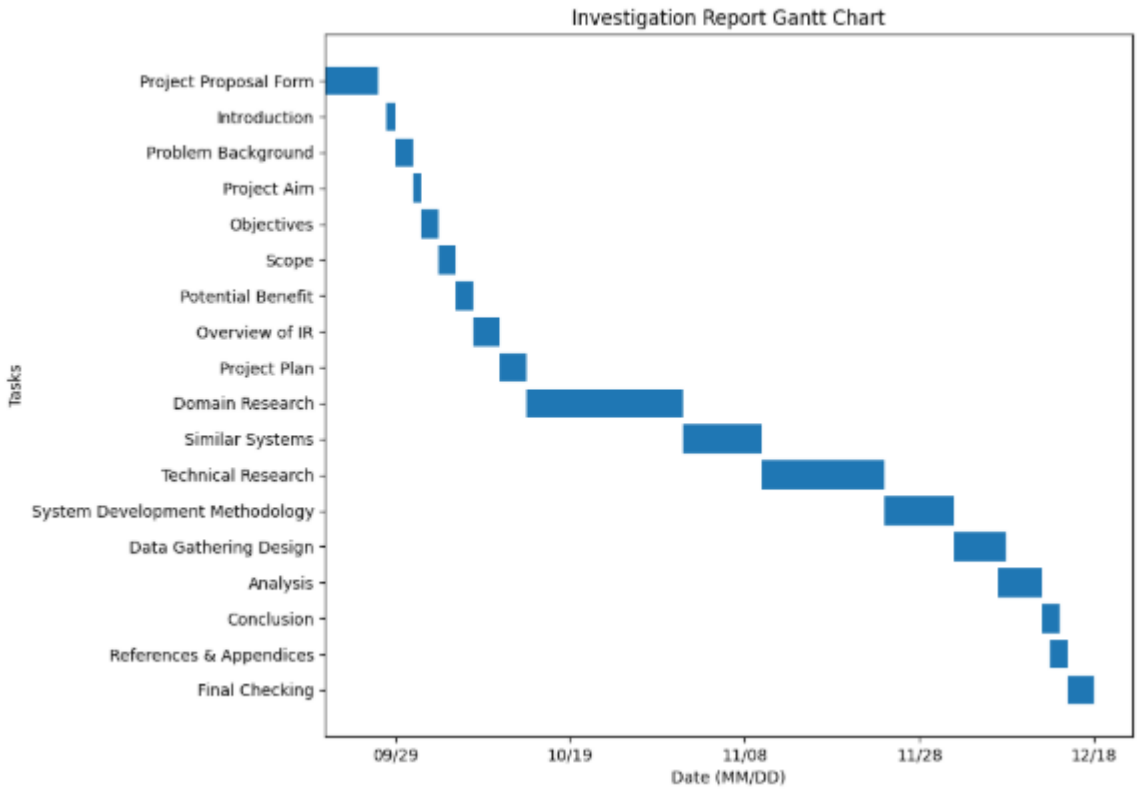


Figure 66: Gantt Chart

Appendix E: Respondence Demographic Profile

Table 6: Demographic Profile

Participant	Age	Educational Background	Gender
Ethan	18-24	Bachelor's Degree	Male
Sophia	18-24	Bachelor's Degree	Female
Mason	18-24	Bachelor's Degree	Male
Isabella	18-24	Bachelor's Degree	Female
Liam	18-24	Bachelor's Degree	Male
Olivia	18-24	Bachelor's Degree	Female
Noah	18-24	Bachelor's Degree	Male
Ava	18-24	Bachelor's Degree	Male
Jacob	18-24	Bachelor's Degree	Male
Emily	18-24	Bachelor's Degree	Female
Lucas	18-24	Bachelor's Degree	Male
Mia	18-24	Bachelor's Degree	Male
William	18-24	Bachelor's Degree	Male
Charlotte	18-24	Bachelor's Degree	Male
James	18-24	Bachelor's Degree	Male
Amelia	18-24	Bachelor's Degree	Female

Henry	18-24	Bachelor's Degree	Male
Harper	18-24	Bachelor's Degree	Male
Michael	18-24	Bachelor's Degree	Male
Evelyn	18-24	Bachelor's Degree	Female
Benjamin	18-24	Bachelor's Degree	Male
Jun	25-34	Bachelor's Degree	Prefer not to say
Joan	25-34	Bachelor's Degree	Prefer not to say
Junna	25-34	Bachelor's Degree	Prefer not to say
Jin	25-34	Bachelor's Degree	Prefer not to say
Jinna	25-34	Diploma/Certificate	Prefer not to say
Jen	25-34	Diploma/Certificate	Prefer not to say
Janne	25-34	Diploma/Certificate	Prefer not to say
Jinne	25-34	Diploma/Certificate	Prefer not to say
Joana	25-34	Diploma/Certificate	Prefer not to say
Bun	25-34	Diploma/Certificate	Prefer not to say
Ben	25-34	Diploma/Certificate	Prefer not to say
Ban	35-44	Diploma/Certificate	Prefer not to say
Banne	35-44	High School/Secondary Education	Prefer not to say

Bunner	35-44	Master's Degree	Prefer not to say
Bendle	45-54	Master's Degree	Prefer not to say
Badle	55 and above	Master's Degree	Prefer not to say